MILITARY SPECIFICATION

FREQUENCY CHANGER, STATIC, AIR COOLED (NAVAL SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers a 150 kilowatt (kW) (at unity power factor) solid state, air cooled frequency converter for the conversion of electrical power from 60 hertz (Hz), 3-phase, to 400 Hz, 3-phase, for non-nuclear power Naval surface ship use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6130

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MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-M-7793	- Meter, Time Totalizing.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-F-15160	- Fuses: Instrument, Power, and Telephone.
MIL-M-16034	- Meters, Electrical-Indicating (Switchboard and Portable Types).
MIL-M-16034/9	- Voltmeters, A.C. and D.C., Portable, 0.25-Percent Accuracy Class.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-S-19500	- Semiconductor Devices, General Specification for.
MIL-C-24643	- Cable and Cord, Electrical, Low Smoke, for Shipboard Use, General Specification for.
MIL-C-24643/18	- Cable, Electrical, 1000 Volts, Type LSMSCU (Including Variations LSMSCA and LSMSCS).
MIL-M-38510	- Microcircuits, General Specification for.

STANDARDS

MILITARY

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-470	- Maintainability Program for Systems and Equipment.
MIL-STD-471	- Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-740-1	- Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.

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MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781 - Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-1310 - Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety.
MIL-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current. (Metric)
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1629 - Procedures for Performing a Failure Mode, Effects and Criticality Analysis.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.
MIL-HDBK-781 - Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government publication. The following other Government publication forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0967-LP-597-1011 - Part Application and Reliability Information Manual for Navy Electronic Equipment.

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
IEEE 200 - Reference Designations for Electrical and Electronics Parts and Equipments. (DoD adopted)

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, NY 10017.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.2 Performance.

3.2.1 Input power. The input to the equipment (see 6.6.1) shall be 440 volts, 60 Hz, 3-phase, 3-wire, ungrounded, type I power complying with the characteristics and ranges as specified in MIL-STD-1399, section 300. Equipment shall not be damaged when the input voltage phase rotation is in a reverse order (AC-CB-BA).

3.2.1.1 Self-protection from input voltage variation and transient. When the equipment is supplying any load from no-load to rated-load, the equipment shall withstand, without damage or blowing of fuses, any input voltage or frequency variation, or both, outside the transient limits specified in MIL-STD-1399, section 300 (see 4.7.5). When the input voltage and frequency exceed the input transient limits, they may take any of the following conditions:

(a) The input voltage instantly drops to zero.
(b) The input voltage drops between 0 and 82 percent of nominal and input frequency drops between 0 and 92 percent of nominal. The voltage-frequency relation during the period shall be such that voltage will be maintained at nominal until the frequency drops down to about 83 percent of nominal, then the voltage drops in proportion to the frequency at a frequency less than 83 percent, and finally the input voltage collapses to zero at a frequency about 35 percent.
(c) The input voltage rises to 150 percent of nominal or input frequency rises to 125 percent of nominal, or both.

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Upon return of the input voltage and frequency to the specified range (see 3.2.1), the equipment shall operate and supply the load with the same output voltage and frequency as that prior to the input power changes. The performance of the equipment shall meet all the requirements of this specification without any adjustment or replacement of parts. If the equipment has shut off, normal starting procedure (see 3.2.1.5) shall be used to restart the equipment.

3.2.1.2 Self-protection from input spikes. When the equipment input voltage experiences several transients (line-to-line or line-to-ground, or both) up to 2500 volts, having a basic impulse level (BIL) waveshape as shown on figure 1, the equipment output voltage waveform shall not deviate from that specified in 3.2.2.8. Equipment shall continue to operate without shutting down or malfunction. The calculated 60 Hz source surge impedance shall be as specified in table I (see 4.7.5).

TABLE I. Calculated 60 Hz source surge impedance.

Navy type cable	Current rating of cable (amperes at 50°C)	Surge impedance (ohms)
T14	47	26
T23	64	26
T50	101	20
T75	136	20
T100	160	19
T150	216	18
T200	250	18
T300	320	18
T400	400	17

3.2.1.3 Input current harmonics. When the equipment is powered by a 60-Hz sinusoidal voltage of the specified range (see 3.2.1) and with the equipment supplying any load from no-load to rated-load, the total current harmonic content shall be not more than 5 percent of the rated-load input current. Individual harmonic currents from the 2nd harmonic through the 32nd harmonic shall be not more than 3 percent. Individual harmonics above the 32nd harmonic until the 333rd harmonic, shall be not more than 100/n percent, where n is the number of the harmonic (see 4.7.6).

3.2.1.4 Input current balance. When the equipment is supplying any load from 10 percent to rated-load and input 3-phase voltages are balanced within 4 volts, the input current shall be balanced to the extent that current in any input line shall not differ by more than 5 percent of the rated-load input current from the arithmetic average current of the three input lines. When the equipment is supplying any load from 15 percent to rated-load and input 3-phase voltages are unbalanced by 3 percent as specified in MIL-STD-1399, section 300, the input current shall be balanced to the extent that the current in any input line shall not differ by more than 10 percent of the rated-load input current from the arithmetic average current of the three input lines (see 4.7.21). The 60 Hz source impedance shall be as specified in table II.

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TABLE II. 60 Hz maximum power source impedance in ohms.1/

Load power factor	1.0		0.8		0.7		0.6		0.5		0.4	
Load kVA 3-phase	R ohms	X ohms	R ohms	X ohms	R ohms	X ohms	R ohms	X ohms	R ohms	X ohms	R ohms	X ohms
10	0.81	0.058	0.97	0.058	1.1	0.058	1.2	0.077	1.4	0.077	1.5	0.077
20	.42	.058	.51	.058	.57	.058	.64	.067	.70	.067	.81	.067
30	.30	.058	.36	.058	.39	.058	.44	.058	.49	.071	.54	.071
40	.23	.058	.28	.058	.29	.058	.33	.058	.37	.067	.42	.067
50	.19	.058	.23	.058	.23	.058	.27	.058	.29	.058	.34	.069
60	.17	.061	.19	.061	.22	.061	.22	.061	.25	.061	.28	.071
70	.15	.060	.17	.060	.18	.069	.20	.069	.22	.069	.24	.069
80	.14	.060	.15	.070	.16	.070	.17	.070	.19	.070	.21	.070
90	.14	.060	.14	.060	.15	.060	.16	.069	.17	.069	.19	.069
100	.12	.060	.13	.060	.14	.069	.15	.069	.16	.069	.17	.069

1/ To convert to per unit values:

$$Xpu = \frac{X \text{ (ohms) (kVA of load) } 1000}{(\text{L-L voltage in volts})^2}$$

3.2.1.5 Starting time. Equipment shall accept any load from no-load to rated-load and operate in compliance with requirements specified herein, within 1 second of command for output power (see 4.7.7). Initial power up of unit may be delayed up to 60 seconds for diagnostic self test.

3.2.1.5.1 Prohibition of start. Equipment shall meet all load requirements when the input voltage is within a range of 390 to 490 volts and the input frequency is within 55 to 65 Hz. Equipment shall be constructed such that it cannot be started when input voltage or input frequency is outside the range where output power can be maintained.

3.2.1.6 Starting input current. Input current shall be limited during the starting time (see 3.2.1.5). The peak input starting (or in-rush) current shall be not greater than 150 percent of the peak full-load steady-state input current (see 4.7.7).

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3.2.1.7 No-load input current. The maximum no-load input current shall be not greater than 10 percent of rated-load input current (see 4.7.19).

3.2.1.8 Grounding. Equipment shall operate as specified herein, when connected to an ungrounded ship's input power system having 3 to 20 microfarad capacitance from any line to ground, or with any single input or output terminal intermittently or permanently grounded. The equipment input or output shall not have any direct grounding or grounding through capacitors (see 4.7.33).

3.2.1.9 Input power factor. The input power factor presented to the 60 Hz input power by the equipment shall be between 0.75 lagging and 0.9 leading.

3.2.2 Output voltage. Rated output voltage shall be 450 volts root mean square (Vrms). The output circuit shall be delta-connected, ungrounded, 3-phase, 3-wire, 400 Hz for all units. The phase rotation of the output voltage shall be a clockwise sequence of AB-BC-CA.

3.2.2.1 Voltage regulation. The average of the Vrms values of the output voltage shall be maintained at the regulated point (the equipment output terminals (see 3.2.2.2)) within plus or minus 0.5 percent of rated output voltage under the following variable conditions and any combination thereof, at all voltage settings within the adjustment range as specified in 3.2.2.3 (see 4.7.8):

(a) A voltage drop from the terminals of the equipment to the regulated point of any amount from 0 to 7 percent of rated output voltage, as specified in 3.2.2.2.
(b) A load variation from no-load to 125 percent of rated-load.
(c) Load power factor varying between 0.7 lagging and 0.9 leading.
(d) An input voltage and frequency variation within the range specified herein.
(e) Variation of the equipment temperature from cold start (at ambient temperature ranging from 0 to 50 degrees Celsius (°C)) to maximum operating temperature (at ambient temperature ranging from 0 to 50°C).
(f) Under the unbalanced load conditions specified in 3.2.2.11.

3.2.2.2 Line drop compensation. The converter shall compensate for a full-load line voltage drop to the load of 7 percent (see 4.7.9). While operating in the line drop compensation mode, the converter shall meet the voltage regulation requirements of 3.2.2.1 at a single point in the distribution system, defined as the load terminals. Other performance requirements of this specification shall be met at the converter output terminals. Line drop compensation shall not be applied to converters operating in parallel.

3.2.2.3 Voltage adjustment. Voltage shall be adjustable not less than plus or minus 5 percent of nominal voltage (427.5 to 472.5 volts). Adjustment shall be operator adjustable within 0.1 percent (0.45 volt) from the front control panel and monitored by digital voltmeter readout (see 3.2.10 and 4.7.10).

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3.2.2.4 Transient output voltage and recovery. The sudden change of 0.8 lagging power factor loads on the equipment from no-load to half-load, from half-load to rated-load, from rated-load to half-load, and from three-fourths load to one-fourth load shall cause an output voltage deviation of not more than plus or minus 5.0 percent of rated voltage. The voltage at the equipment output terminals shall recover and stay within the steady state regulation band (see 3.2.2.1), within 2.5 milliseconds. Circuits shall be so constructed that voltage transients developed within the equipment shall not reflect back into the input power system such that transient limits as specified in MIL-STD-1399, section 300, are exceeded for alternating current (ac) input (type I power) (see 4.7.11).

3.2.2.4.1 Steady-state nonlinear loads. Typical steady-state nonlinear loads shall be defined as loads developed by the 3-phase full-wave rectifying systems, and shall have maximum harmonic current content imposed on the equipment output line in accordance with the following formula, when the load system is supplying resistive loads:

$$I_N = \frac{100}{N} \text{ percent of fundamental current}$$

I is the Nth harmonic current and N is the harmonic number. N shall not be any even number or multiple of three. When a nonlinear load up to 70 percent of the rated load is applied to the equipment, the output voltage waveform shall remain within the limits specified in 3.2.2.8, the output voltage regulation shall be in accordance with 3.2.2.1, and the voltage amplitude modulation shall be in accordance with 3.2.2.7.

3.2.2.5 Frequency regulation. Output frequency shall be maintained within plus or minus 0.5 percent of rated frequency under the following conditions or combination thereof (see 4.7.8):

(a) A load variation from no-load to 125 percent of rated-load.
(b) Load power factor varying between 0.7 lagging and 0.9 leading.
(c) An input voltage and frequency variation within the range specified herein.
(d) Variation of the equipment temperature from cold start (at ambient temperature ranging from 0 to 50°C) to maximum operating temperature (at ambient temperature ranging from 0 to 50°C).
(e) Variation of output voltage specified in 3.2.2.3.

3.2.2.6 Transient frequency deviation and recovery. A sudden change of 0.8 lagging power factor loads on the equipment from no-load to rated-load and from rated-load to no-load shall not cause a frequency deviation greater than the steady-state band specified in 3.2.2.5 (see 4.7.11).

3.2.2.7 Amplitude and frequency modulation. Amplitude and frequency modulation of the output voltage (as specified in MIL-STD-1399, section 300) shall not exceed 1 percent and 0.5 percent, respectively, under any of the conditions or combinations of conditions as specified in 3.2.2.5 (see 4.7.12).

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3.2.2.8 Output voltage waveform. Under any load conditions, between no-load and rated-load, the root mean square (rms) value of all of the harmonics in the output voltage shall not exceed 3 percent of the rated output voltage. The rms value of any single harmonic shall not exceed 2 percent of the rms value of the rated output voltage. The deviation factor (as specified in MIL-STD-1399, section 300) of the output voltage waveform shall not exceed 5 percent. Peak-to-peak amplitude of voltage spikes (between the phases) of any frequencies shall not exceed 3 percent of the rms value of the rated output voltage (see 4.7.13).

3.2.2.9 Output phase displacement. Displacement angle between adjacent voltage phases shall be 120 $\pm$ 1 degrees under all conditions of operation specified herein (see 4.7.14).

3.2.2.10 Output voltage balance under balanced load conditions. The difference between phase voltage at the voltage regulated point of the equipment shall be not greater than 0.5 percent of rated voltage at no-load and at any balanced 3-phase load up to rated current with any power factor between 0.7 lagging and 0.9 leading (see 4.7.15).

3.2.2.11 Output voltage balance under unbalanced load conditions. The voltage variation between phases at the equipment output terminals shall be not greater than 1 percent of the rated voltage under the unbalanced load conditions as shown in table III, when the output voltage is regulated at the equipment output terminals (see 4.7.16).

TABLE III. Unbalanced loads.

Load conditions	Line A (percent of rated current)	Line B (percent of rated current)	Line C (percent of rated current)	Power factor
1	90	75	75	0.7 lagging to unity
2	75	90	75	0.7 lagging to unity
3	75	75	90	0.7 lagging to unity

Equipment shall operate with a single phase 0.7 power factor load carrying a current 15 percent of rated current. The voltage of the loaded phase at the voltage regulated point shall have regulation as specified in 3.2.2.1.

3.2.2.12 Input voltage change effect. When the equipment is supplying any load from no-load to rated-load and with any output and input steady-state voltage and frequency within the range specified in 3.2.1, the occurrence of any input transient voltage up to plus or minus 20 percent of nominal, with a recovery time of 2 seconds to the set limits, as specified for type I power in MIL-STD-1399, section 300, shall not cause an output voltage deviation greater than plus or minus 3 percent of the initial value. The output voltage shall recover to and remain within the steady-state regulation band within 0.02 second from the termination of the input voltage change (see 4.7.18).

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3.2.3 Power ratings. The continuous power ratings of the 150 kilovolt-ampere (kVA) unit shall be 150 kW at unity power factor and 150 kVA from 0.9 leading through 0.7 lagging power factor (see 4.7.20). The power range of the unit is defined on figure 2.

3.2.3.1 Duty cycle. Equipment shall operate continuously as specified in 3.2.3.

3.2.4 Efficiency. The overall efficiency of the equipment shall be not less than 87 percent when operating at rated input voltage, frequency, load and power factor (see 4.7.19).

3.2.5 No-load losses. No-load losses of equipment shall be not more than 60 percent of the rated-load losses at nominal input voltage and frequency (see 4.7.19).

3.2.6 Overload. Equipment shall provide and sustain an overload current of 125 percent of rated current at the rated output voltage for not less than 30 minutes and meet all requirements specified herein (see 4.7.22). The output voltage regulation for any overload current up to 125 percent of the rated output current shall be as specified in 3.2.2.1.

3.2.7 Fault and malfunction protection.

3.2.7.1 Fault isolation. Equipment shall incorporate fault isolation for each module or drawer (see 6.6.4 and 6.6.5) to prevent cascading to other modules or drawers. A shorted or non-commutated power semiconductor shall not result in damage to any other part of the equipment, except that fuses may blow (see 6.6.6).

3.2.7.2 Fault current. Fault currents shall be defined as output current between 125 and 300 percent of rated current. The 150 kVA unit shall provide and sustain any fault current up to 300 percent while maintaining output voltage within the requirements of MIL-STD-1399, section 300, type III. Such fault currents will be provided and sustained for the following time periods (see 4.7.23):

(a) Between 210 and 300 percent, 0.1 second minimum
(b) Between 130 and 210 percent, 2 seconds minimum
(c) Between 100 and 130 percent, 30 minutes minimum

In the event the fault current exceeds the magnitudes or durations listed, the 150 kVA unit shall operate in current limit mode at 130 percent of rated current for not less than 15 minutes or until the fault current reduces to below 130 percent of rated current. Fault current performance is defined on figures 3 and 4.

3.2.7.3 Short circuit protection. A short circuit shall be defined as an impedance which requires a power level greater than 300 percent of nominal output power or when the output current exceeds 576 amperes. On the occurrence of a short circuit, the 150 kVA unit shall revert to current limit mode within 50 milliseconds (ms) at 130 percent of nominal current (see figure 3 and 4.7.23). Indication shall be provided to show that a short circuit occurred (see 3.2.10).

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3.2.7.4 Internal circuit protection. Fuses shall be provided for power circuits and semiconductors as backup for the current limiting protection (see 3.2.7.3). Each drawer containing power switching semiconductors shall be separately fused. Circuits with filter capacitors shall have fuse protection against capacitor failure.

3.2.7.5 Protection for utilization equipment. A protective scheme shall be provided to shut down the equipment if the output voltage shown in table IV exists. The equipment shutdown shall be delayed from not less than 45 ms to not greater than 250 ms to prevent nuisance tripping because of possible system transients (see 4.7.17). An alarm indicator shall record the over-voltage condition. In addition, an alarm indicator shall record an output under-voltage condition of 425 volts or less. Both alarm indicators shall latch until manually reset.

TABLE IV. Shutdown voltage setting.

Condition	Setting (percent of rated output)	Tolerance (percent of rated output)
Over-voltage	110	Plus 4 minus 0

3.2.8 Parallel operation. Equipment shall provide paralleling operation capability at the levels of reliability and maintainability specified herein. It shall be possible to parallel up to a maximum of 6 units, without interrupting the load and without introducing transient voltages at the input line and output load bus outside the specified limits. After the units are connected for parallel operation and the output voltage of each unit has been manually adjusted (see 3.2.2.3) so that the maximum terminal voltage difference between the units is not more than 1 volt, the synchronization and parallel operation of the units shall be automatically accomplished to meet all requirements specified herein. The units shall parallel under any load conditions from no-load to rated-load. Loss of any individual frequency converter control shall not cause a non-synchronous condition (see 4.7.24).

3.2.8.1 Synchronization. After the units are connected for parallel operation, synchronization shall take place automatically, with no time lag after connection.

3.2.8.2 Real and reactive load division. When paralleling identical size units, the maximum difference in kW supplied by the units in parallel shall not exceed 7.5 percent of the kW rating of one of the units under load conditions between 10 and 100 percent of the total rated-load of the paralleled units. The maximum difference in kilovar (kvar) supplied by the units operating in parallel shall not exceed 10 percent of the kvar rating of one of the units under any load condition between 10 and 100 percent of the total rated-load of the paralleled units. When paralleling different size units, the output of each unit shall be directly proportioned to the kW rating of the individual unit. The maximum

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difference in kW between the actual kW supplied by each unit and its proportional share of the total kW shall not exceed plus or minus 5 percent of the average kW rating of units in parallel under load conditions between 10 and 100 percent of the total rated load of the paralleled units. The maximum difference in kvar between the actual kvar supplied by each unit and its proportional share of the total kvar shall not exceed plus or minus 5 percent of the average kvar rating of units in parallel under load conditions between 10 and 100 percent of the total rated load of the paralleled units. The above requirements shall apply for input voltage variations specified in 3.2.1.1 and other environmental requirements specified herein.

3.2.8.3 Selective tripping. Equipment shall operate so that when any malfunction occurs to any one of the units operated in parallel, the malfunctioned unit shall automatically shut itself down and the remaining units shall continue to operate in parallel, as specified herein.

3.2.8.4 Arrangement of parallel operation. The parallel operation of up to the maximum six units shall be connected to a common bus (see 3.2.8.4.1) or to a split bus (see 3.2.8.4.2) arrangement. Paralleling requirements as specified in 3.2.8, 3.2.8.1, 3.2.8.2, and 3.2.8.3 shall apply to both the common bus and the split bus arrangements.

3.2.8.4.1 Common bus parallel operation. The common bus parallel operation shall be connected as shown on figure 5 (figure 5 only shows two units). When the 400 Hz bus is located in the same compartment with the frequency changers such that the impedance of the 400-Hz cable between the units and the 400-Hz bus is considered negligible, the voltage of the common bus shall meet all output requirements specified herein (see 3.2.2 through 3.2.2.12).

3.2.8.4.2 Split bus parallel operation. The split bus parallel operation shall be connected as shown on figure 6 (figure 6 shows two units and two buses) up to the maximum of six units and three buses in any combination. When each unit is located close to its own 400-Hz bus such that the impedance of the 400-Hz cable between the unit and its bus is considered negligible and the maximum impedance of each bus tie cable is less than 2 percent (based on the rated output current of one unit), the voltage at one of the 400-Hz buses as A, B or C (bus C is not shown on figure 6) shall meet all output requirements specified herein (see 3.2.2 through 3.2.2.12, 3.2.8.1 and 3.2.8.3). Selection of the bus to meet the output voltage requirements may be accomplished by a control cable connecting between the unit and its bus. One normally-open and one normally-closed auxiliary contact are available in each bus tie breaker and can be used to provide bus tie breaker status information, for the control of the split bus parallel operation. In this operation mode, if input power (60 Hz) is lost on one bus, this shall not cause units not fed from that input bus to shut down even if 400-Hz bus tie breakers are closed. Operation of bus tie breakers shall not cause improper operation of any units on line.

3.2.9 Degraded operation. The 150-kVA unit shall be constructed with operationally redundant modules. The operational failure of one rectifier module or one inverter module shall not inhibit operation of the unit or cause operation which will result in an output which does not meet the power quality requirements

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of MIL-STD-1399, section 300, type III. Visual indication shall be provided to indicate the failed module and degraded mode of operation (see 3.2.10). When operating in a degraded mode, the unit shall provide 125 kVA power continually. Transient overload capability shall be reduced to 150 kVA for 2 seconds and the unit shall limit current at not less than 160 amperes per phase.

3.2.9.1 While operating in a degraded mode in parallel operation, the unit shall operate at a nominal power level of 75 kVA and sharing with other units shall be limited to one-half normal load share.

3.2.9.2 In the event that an inverter or rectifier module sustains damage to its local control circuitry, this failure shall not affect rectifier or inverter module operation. Failure in control circuits shall not cause failure of power modules.

3.2.10 Status and metering indications. The unit shall provide all system indications and operational instructions by means of an electroluminous screen in a digital and text format located on the front of the unit.

3.3 Construction.

3.3.1 General requirements. In addition to the requirements of this specification, the equipment shall be in accordance with MIL-E-917 except the following (see 6.3):

(a) Fundamental equipment characteristics (excluding safety)
(b) Grounding
(c) Input power variations (ac equipment)
(d) Inclined operation
(e) Ambient temperature
(f) Impact shock and vibration
(g) Equipment mounting
(h) Radio interference
(i) Noise reduction for submarine
(j) Piping system
(k) Workmanship

3.3.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specifically specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

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3.3.3 Electrical construction. Equipment shall utilize block modules concept with solid state circuitry and devices only. Electromechanical type relays may be used only under the conditions applicable to the use of nonstandard parts (see 3.4.1). Vacuum tubes or other similar thermionic devices shall not be used. Control and logic circuits shall have a common negative. Control wires shall be separated from the power conductors to avoid noise pickup in the control circuits. Electrical isolation between control circuits and power semiconductor devices by optical coupling may be used.

3.3.3.1 Isolation. Equipment shall be electrically isolated from the equipment output. Isolation transformers shall have electrically isolated windings (primary and secondary windings). Auto-transformers shall not be used.

3.3.3.2 Input and output transformers. Equipment input and output transformers shall be a balanced arrangement (for example: 3-phase delta connection, 3-phase wye connection, 6-phase star connection, and so forth). Scott "T" and open-delta connections shall not be used.

3.3.3.3 Dielectric withstanding voltage and insulation resistance. Insulation resistance of the circuits shall be not less than 10 megohms when measured with units at ambient temperature of 24 ± 2°C (see 4.7.2). Dielectric strength shall be twice rated plus 1000 volts for 60 seconds (see 4.7.3). The equipment (except the control drawer) shall be such that required insulation resistance and dielectric withstanding voltage tests on the insulation system can be conducted without either incipient or catastrophic damage or without significant (within acceptable product variability tolerance) degradation to any material or part, including the electrical insulation. Furthermore, a defect in the insulation system shall not cause either of the tests, when properly conducted, to result in damage or degradation to any other material or part except that which possibly may be inflicted to the immediately adjacent parts separated by the defective insulation.

3.3.4 Size and weight. The equipment, including air plenums, shall not exceed the weight and dimensions shown in table V.

TABLE V. Weight and dimensions.

Power rating	Dimensions			Maximum weight per unit (pounds)
	Width (inches)	Depth (inches)	Height (inches)	
150 kVA	38	36	79	4600

Weight of equipment shall be kept to a minimum. At delivery, the net weight and also the operating weight shall be indicated. In addition, the vertical, longitudinal, and transverse location of the center of gravity shall be shown on the outline or assembly drawing.

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3.3.5 Cabinet enclosure. Equipment shall have a drip-proof protected (45-degree) enclosure as specified in MIL-STD-108 (see 4.7.4).

3.3.6 Ground connection. Exposed metal parts of the cabinets shall be at ground (ship's hull) potential at all times. Exposed metal portion of electrical parts (such as switches and rheostats) which are not part of the electrical circuits shall be electrically grounded to the frame of the cabinet. A ground stud shall be provided on each cabinet as specified in MIL-STD-1310 to accept a class C, type II bond strap which will be supplied and installed by the contracting activity. The ground stub shall be clearly marked and shall be readily accessible and located or protected to reduce the probability of breakage during transit or operation.

3.3.7 Cabinet mounted control. An elapsed time meter and the output voltage adjustment device (see 3.2.2.3) shall be mounted on the cabinet (not on a drawer front panel). The elapsed time meter shall have a minimum readout of 00000 hours and shall be as specified in MIL-M-7793.

3.3.8 Remote control. Signals for the remote control, indication and interlock shall be provided on a remote control panel for single unit and parallel operation. The remote control panel shall be connected to the unit by a type LSMSCV cable, as specified in MIL-C-24643. The cable will be furnished by the installation activity. The maximum length of the interconnection cable between the equipment and the remote control panel shall be 500 feet. The controls and indicators on the remote control panel shall be a computer terminal or as follows:

(a) Pushbutton or switch for unit start up and shut down.
(b) Pushbutton or switch for closing and opening the output circuit breaker.
(c) Indication light to show that the unit is energized.
(d) Indication light to show that the output circuit breaker is closed.
(e) Output current ammeter.

The frequency changer shall provide signals for (a) and (c). Controls and indicators as shown on figures 5 and 6 shall be located on the remote control panel. The installation activity will supply the remote control panel, complete with controls and indicators wired up, installed (including the interconnection cable) and ready for connection to the frequency changer. Remote control of the unit should be defeated when control of the unit is local.

3.3.9 Identification of equipment divisions. To provide identification that will aid documentation and to facilitate maintenance, each part, module, drawer or cabinet, and unit of the equipment shall be identified by reference designations in accordance with IEEE 200 and with 3.3.9.1 and 3.3.9.2.

3.3.9.1 Identification of parts leads. For each polarized part, such as electrolytic capacitors and rectifiers, the polarity identification shall be marked on the part mounting surface in order to help assure proper replacement of the part. For multiple lead parts such as transistors where proper circuit operation is dependent upon a specific lead hookup, appropriate lead identification shall be marked on the part mounting surface to help assure proper replacement of the part.

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3.3.9.2 Method of marking. Markings shall be permanent and legible. The markings on plastic or metallic materials shall be made by stamping, engraving, stenciling or rubber stamping with smudgeproof ink covered with a coat of clear lacquer or silk screening. Decalcomanias or paper labels shall not be used.

3.3.9.3 Identification plate. Equipment shall be equipped with an identification plate. The identification plate shall be metallic of types A, B, C or H as specified in MIL-P-15024 and MIL-P-15024/5.

3.3.9.4 Operation instruction plate. Equipment shall be equipped with an operation instruction plate. The plate shall be as specified in MIL-P-15024 and MIL-P-15024/5, types A, B, C or H (metallic only).

3.3.10 Human engineering. Human engineering shall be as specified in MIL-STD-1472.

3.3.11 Safety. The equipment shall be as specified in MIL-E-917. Personnel shall be protected from capacitor discharges as specified in requirement 1 of MIL-STD-454. When the equipment input isolation circuit breaker (supplied by installation activity) is in the off position, the output isolation circuit breaker (supplied by installation activity) and all circuits of the equipment shall be de-energized (only the input terminals of the input circuit breaker and the output terminals of the output circuit breaker, and their associated controls may be energized).

3.4 Component requirements.

3.4.1 Selection of parts. The parts, modules, and drawers which perform similar functions shall be standardized and interchangeable to the maximum extent practicable. The number of different parts used in the equipment shall be kept to a minimum consistent with its intended use. Selection of parts shall be as specified in MIL-E-917. Nonstandard parts shall not be incorporated into the equipment without prior approval from NAVSEA.

3.4.1.1 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.4.2 Contractor furnished parts. Supplementing other requirements for parts identification, contractor furnished parts shall be identified in parts lists and drawings by part manufacturer's name and part number in addition to the equipment manufacturer's part number.

3.4.3 Fuses. Fuses used in series with power semiconductors shall be the type for protection of semiconductor devices. Fuses shall be provided for power circuits and semiconductors as backup for the current limiting protection (see 3.2.7.4). Except for semiconductor protection, fuses shall be as specified in MIL-F-15160. Fuse voltage rating shall be not less than the maximum circuit application voltage.

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3.4.4 Solid state circuitry.

3.4.4.1 Thyristor firing circuit. Identical thyristor firing circuit modules shall be used throughout the equipment to the maximum extent possible. Firing circuits shall be constructed for maximum noise immunity utilizing decoupling, shielding, low impedance, twisted pair cables, reverse bias and proper wiring layout to maximum extent possible.

3.4.4.2 Electrical control circuits. Established reliability components shall be used in the logic circuitry. Microcircuits shall be as specified in MIL-M-38510, class B or screened as specified in MIL-STD-883. Semiconductors shall be as specified in MIL-S-19500, level JANTX. Resistors and capacitors shall be as specified in applicable military specifications.

3.4.5 Semiconductor devices. Semiconductor devices shall be chosen and applied as specified in MIL-E-917 and as specified in 3.4.5.1 through 3.4.5.7. Unless otherwise specified (see 6.2), all semiconductor devices shall be silicon. Discrete semiconductors shall be as specified in TX requirements of MIL-S-19500 and the applicable detailed specification, where such semiconductors are available.

3.4.5.1 Standardization promotion. When a justifiable need exists for a device not covered by an existing military specification but which can be provided by extension of a series or family of military devices, such extension shall be employed in preference to choice of a device from a new family or series. The power semiconductor devices used (diodes and semiconductor control rectifier (SCR's)) shall be presently available without further development. The power semiconductors shall have not less than two sources of supply, where possible.

3.4.5.2 Device selection prohibition. The equipment shall meet all requirements without the need for device selection beyond that covered by the applicable device detail specification. Semiconductor devices of the same type designation shall be interchangeable for all equipment without any restrictions, including source of manufacture.

3.4.5.3 Reverse polarity prohibition. Devices of standard polarity shall be used wherever feasible. Devices of reverse polarity may be used only under the conditions applicable to the use of nonstandard parts (see 3.4.1).

3.4.5.4 Stress factors and limits. Circuits shall be constructed and devices shall be applied so that the stress limits as specified in table VI will not be exceeded under any condition of operation, standby and during tests specified for the equipment (see 4.7.36). The values specified are limits and not design factors nor design margins. The contractor shall determine and provide sufficient margin between the applied stresses and the specified limits to ensure that the required life and reliability are achieved. In no case shall the relative values of the steady state voltage variation, over-voltage, transient voltage and surge voltage spikes, for the circuit under consideration, be less than the values specified in 3.2.1, 3.2.1.1 and 3.2.1.2 except where the equipment includes features to limit or suppress over-voltage and surge voltage to lower values. When such exception is used, the contractor shall provide a

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detailed description of the features and how the circuit functions to limit peak over-voltage and surge voltage impressed on the semiconductor devices to whatever lower values apply. Circuits which employ controlled rectifiers or transistors shall suppress surge voltages to less than three times the supply nominal rms voltage. Semiconductor devices shall be applied in accordance with 3.4.5.5. Surge voltage suppressors shall be in accordance with 3.4.5.6. Heat dissipaters (heat sinks, cooling fins) shall be in accordance with 3.4.5.7.

TABLE VI. Application stress limits and rerating factors.

Rating multiplier (RM)				
	Stress limits -			
Stress	RM lower	Rating upper	Rating symbol	Rating
Semiconductor devices				
A. General:				
1. Equipment design ambient, T(AE)				
a. T(AE, min)	---	---		0°C
b. T(AE, max)	---	---		50°C
2. Component design ambient, T(AC)				
a. T(AC, min)	---	1.0		T(AE, min)
b. T(AC, max)	1.0	---		T(AE,max)+T(AC)-T(AE) T(AC)-T(AE) is the maximum calculated temperature difference between the component ambient and the equipment ambient at the operating conditions being evaluated.
B. Rectifiers, silicon controlled rectifier (SCR) and gate turn-off thyristor (GTO):				
1. Dc reserve blocking voltage	---	0.4	V(RRM)	Repetitive peak reverse voltage
2. Nominal working peak reverse voltage	---	0.5	V(RRM)	Repetitive peak reverse voltage

than the values specified in 3.2.1, 3.2.1.1 and 3.2.1.2 except where the equipment includes features to limit or suppress over-voltage and surge voltage to lower values. When such exception is used, the contractor shall provide a

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TABLE VI. Application stress limits and rerating factors - Continued.

Stress	Rating multiplier (RM) Stress limits - RM lower	Rating upper	Rating symbol	Rating
4. Non-repetitive peak reverse voltage	---	1.0	V(RRM)	Repetitive peak reverse voltage
5. Dc forward blocking voltage	---	0.4	V(DRM)	Repetitive peak off-state voltage
6. Nominal working peak forward blocking voltage	---	0.5	V(DRM)	Repetitive peak off-state voltage
7. Repetitive peak forward blocking voltage	---	0.7	V(DRM)	Repetitive off-state voltage
8. Non-repetitive peak forward blocking voltage	---	0.8	V(DRM)	Repetitive peak off-state voltage
9. Maximum rate of rise forward blocking voltage	---	0.5	dv/dt (critical)	Critical rate of rise of forward blocking voltage
10. Minimum duration of voltage reversal (SCR)	2.0	---	t(q)	Turn-off time at maximum rated junction temperature T(j,max)
11. Reapplied forward blocking voltage (SCR)	---	0.7	V(DRM, reapplied)	Reapplied forward blocking voltage at rated t(q) and T(j,max)
12. Rate of rise of reapplied forward blocking voltage (SCR)	---	0.7	dv/dt (reapplied)	Reapplied rate of rise of forward blocking voltage at rate t(q) and T(j,max)
13. Minimum duration of gate turn-off signal (GTO)	2.0	---	t(gq)	Gate controlled turn-off time at maximum rated junction temperature T(j,max)
14. Average forward current:				
a. Nominal value	---	0.7	I(T,AV)	Average forward current at T(j,max) for the application waveform, conduction angle duty cycle and frequency
b. Maximum value	---	1.0	I(T,AV)	

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TABLE VI. Application stress limits and rerating factors - Continued.

Stress	Rating multiplier (RM) Stress limits - RM lower	Rating upper	Rating symbol	Rating
15. Repetitive peak forward current:				
a. Nominal value	---	0.7	I(TRM)	Repetitive peak current at T(j,max) for the application waveform, conduction angle, duty cycle and frequency
b. Maximum value	---	1.0	I(TRM)	
16. Surge current:				
a. Peak, multiple cycle	---	0.7	I(TSM)	Peak surge current, peak half cycle (1/120sec) surge, non-repetitive
b. Peak multiple cycle	---	0.7	I(TSM, multiple cycle)	Peak surge current, rated for the number of cycles expected in the application
c. Peak, sub-cycle	---	0.7	I(TSM, sub-cycle)	Peak surge current rated for the application pulse base width
17. Maximum rate of rise of current during turn-on:				
a. Repetitive maximum	---	0.8	di/dt(REP)	Maximum repetitive rate of rise of current during turn-on at T(j,max) and at rated repetitive peak current for the application waveform conduction angle, duty cycle and frequency
b. Non-repetitive maximum	---	0.5	di/dt (critical)	Critical rate of rise of current, repetitive
c. During surge	---	1.0	di/dt (critical)	Critical rate of rise of current, repetitive

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TABLE VI. Application stress limits and rerating factors - Continued.

Stress	Rating multiplier (RM) Stress limits – RM lower	Rating upper	Rating symbol	Rating
18. Turn-on gate power supply characteristics (SCR and GTO):				
a. Source impedance	---	1.0	R[SG]	Gate power supply characteristics at rated critical rate of the rise of current and peak during turn-on, repetitive
b. Open circuit voltage	1.0	---	V[SG]	
c. Open circuit voltage transition time	---	1.0	t[r,SG]	
19. Turn-off gate power supply characteristics (GTO):				
a. Source impedance	---	1.0	R[SG]	Gate power supply characteristics at rated rate of fall of anode current and peak anode current during turn-off
b. Open circuit voltage	1.0	---	V[SG]	
c. Open circuit voltage transition time	---	1.0	t[f,SG]	
20. Peak reverse gate voltage:				
a. Maximum (SCR & GTO)	---	0.7	V[GRM]	Peak reverse gate voltage
b. Minimum for turn-off (GTO)	2.0	---	V(GQM]	Reverse gate voltage, minimum to cause turn-off
21. Peak reverse gate current (GTO):				
a. Maximum	---	0.7	I[GRM]	Peak reverse gate current
b. Minimum for turn-off	2.0	---	I[GQM]	Reverse gate current, minimum to cause turn-off
22. Peak forward gate current (SCR and GTO):				
a. Maximum for firing	---	0.7	I[GM]	Peak forward gate current
b. Minimum for firing	2.0	---	I[GTM]	Forward gate current, minimum to cause firing
c. Maximum during forward blocking	---	0.0	I[GTM]	
23. Peak forward gate voltage (SCR & GTO):				
a. Maximum for firing	---	0.7	V[GM]	Peak forward gate voltage

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TABLE VI. Application stress limits and rerating factors - Continued.

Stress	Rating multiplier (RM) Stress limits - RM lower	Rating upper	Rating symbol	Rating
b. Minimum for firing	2.0	---	V[GTM]	Forward gate voltage, minimum to cause firing (maximum value)
c. Maximum during forward blocking	---	0.0	V[GTM]	
24. Maximum gate power dissipation (SCR & GTO):				
a. Average	---	0.7	P[G,AV]	Average gate power dissipation
b. Peak	---	1.0	P[GM]	Peak gate power dissipation
25. Junction temperature (calculated):				
a. Average	---	---	T[j,AV]	T[j,max] - 40°C
b. Repetitive peak	---	---	T[j, REP PK]	T[J,max] - 25°C
C. Voltage reference diodes:				
1. Bias current	0.8	1.2		Zener test current (value for rated Zener voltage)
2. Diode reverse current (Zener current):				
a. Minimum	0.1	---		Zener test current (value for rated Zener) voltage
b. Maximum	---	0.5		Maximum Zener current
c. Maximum, continuous duty	---	0.3		Maximum Zener current
3. Junction temperature (calculated)	---	100C		
D. Voltage regulator diodes:				
1. Diode reverse current (Zener current):				
a. Minimum	0.1	---		Zener test current (value for rated Zener voltage)
b. Maximum, steady state	---	0.5		Maximum Zener current
c. Maximum, continuous duty	---	0.3		Maximum Zener current
d. Surge	---	0.5		Surge current

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TABLE VI. Application stress limits and rerating factors - Continued.

Stress	Rating multiplier (RM) Stress limits – RM lower	Rating upper	Rating symbol	Rating
2. Junction temperature (calculated)				
a. Average	---	---	T[j,AV]	T[j,max] - 50°C
b. Peak	---	---	T[j,PK]	T[j,PK] - 25°C
E. Bipolar transistor:				
1. Collector to emitter voltage, cutoff or inverse mode:				
a. Dc	---	0.4	V[CEO]	Collector - emitter voltage
b. Nominal working	---	0.5	V[CEO]	Collector - emitter voltage
c. Repetitive peak	---	0.7	V[CEO]	Collector - emitter voltage
d. Non-repetitive	---	1.0	V[CEO]	Collector - emitter voltage
2. Collector to base voltage, cutoff or inverse mode:				
a. Dc	---	0.4	V[CBO]	Collector to base voltage
b. Nominal working	---	0.5	V[CBO]	Collector to base voltage
c. Repetitive peak	---	0.7	V[CBO]	Collector to base voltage
d. Non-repetitive	---	1.0	V[CBO]	Collector to base voltage
3. Emitter to base voltage, cutoff or inverse mode:				
a. Dc	---	0.4	V[EBO]	Emitter to base voltage
b. Nominal working	---	0.5	V[EBO]	Emitter to base voltage
c. Repetitive peak	---	0.7	V[EBO]	Emitter to base voltage
d. Non-repetitive	---	1.0	V[EBO]	Emitter to base voltage
4. Peak collector current, cutoff of inverse mode	---	0.7	I[EBO]	Emitter cutoff current
5. Peak emitter current, cutoff or inverse mode	---	0.7	I[EBO]	Emitter cutoff current
6. Average collector current, active or saturation mode:				
a. Nominal value	---	0.6	I[C,AV]	Average collector

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TABLE VI. Application stress limits and rerating factors - Continued.

Stress	Rating multiplier (RM) Stress limits – RM lower	Rating upper	Rating symbol	Rating
b. Maximum value	---	0.6	I[C,AV]	Current at rated T[j,max] for the application waveform, conduction angle or period, duty cycle and frequency
7. Repetitive peak collector current, active or saturation mode:				
a. Nominal value	---	0.6	I[C,REP]	Repetitive peak collector current at rated T[j,max] for the application waveform, conduction angle or period, duty angle and frequency
b. Maximum value	---	0.8	I[C,REP]	
8. Non-repetitive collector current active	---	1.0	I[C,REP]	Repetitive peak collector current
9. Peak base current	---	0.7	I[B]	Maximum base current at rated T[j,max]
10. Peak emitter current	---	0.7	I[E]	Maximum emitter current at rated T[j,max]
11. Maximum collector to emitter voltage, active or saturation mode	---	0.7	V[CE]	Repetitive peak collector to emitter voltage at rated repetitive peak collector current and rated T[j,max] for the application waveform, conduction angle or period, duty cycle and frequency
12. Circuit current gain	---	0.5	H[FE]	Static forward current transfer ratio
13. Minimum duration of base controlled turn-off signal	2.0	---	(t[s] + t[f])	Turn-off or cutoff time (storage time and fall time) during turn-off at rated T[j,max]
14. Junction temperature (calculated)				
a. Average	---	---	T[j,AV]	T[j,max] - 50°C
b. Repetitive peak	---	---	T[j,REP PK]	T[j,max] - 25°C

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3.4.5.5 Application limitations. The following limitations and guides shall apply in the application of semiconductor devices:

(a) Devices constructed and rated for conduction cooling shall be mounted on metallic heat dissipaters. Attention shall be given to the compatibility of materials and to the finish coatings on parts mounted together in an assembly. Connections and joints in both electrical and thermal circuits shall be constructed to ensure that their effectiveness will be maintained for the life of the equipment under adverse shipboard conditions, including shock, vibration, thermal cycling, high humidity and salt-laden atmosphere. Means for maintaining positive contact pressure on bolted or clamped joints, under all service conditions, taking into account the aging (including creep, cold flow, relaxation, fatigue, shrinkage and so forth) of materials shall be included.

(b) Direct mounting of devices to their heat dissipater with electrical insulation of the heat dissipaters, as necessary, is preferred to the use of intervening electrical insulating material between the devices and their heat dissipaters. This preferred arrangement shall be used except where it is impracticable to insulate the heat dissipaters, or where other technical considerations make electrical insulation of the semiconductor device from its mounting surface necessary.

(c) Semiconductor devices shall not be operated in parallel to provide current capacity.

(d) Semiconductor devices (diodes, rectifiers, SCR's and transistors) shall not be operated in series. Transistors shall not be used in series to obtain higher voltage capacity.

(e) Voltage reference diodes and voltage regulator diodes shall not be used in parallel to obtain higher current capacity.

3.4.5.6 Surge voltage suppression. Circuits containing devices which do not have inherent capability of withstanding voltage surges of the level as specified in 3.4.5.4 without exceeding allowable stresses shall be protected by incorporating adequate surge voltage suppression within the circuit. The internal generation of voltage surges shall be minimized with respect to both frequency of occurrence and amplitude. Means shall be provided within the circuit both for suppressing any remaining voltage surges generated within the circuit that would otherwise be excessive and for preventing external propagation at a peak level greater than three times the rms voltage for the ac output circuit and that permitted as specified in 3.2.1, 3.2.1.1 and 3.2.1.2 for the input circuit. Surge voltage suppressors shall be in accordance with MIL-E-917.

3.4.5.7 Heat dissipaters. Heat dissipaters for semiconductor devices shall be constructed of metallic material which is either corrosion-resistant, or which is treated or coated to resist corrosion under shipboard environmental conditions. Surfaces which contact the semiconductor device, and surfaces to which electrical contacts are made shall not be painted or anodized, but shall be plated or otherwise coated to form surfaces to which the connections or junctions are made and which will retain their mechanical, thermal and electrical effectiveness for the life of the equipment. The method of processing and the

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preparation of the connections and joints shall be as specified on the applicable heat dissipater drawing. Heat dissipaters shall be sized to dissipate the heat generated by the semiconductor device of highest losses that may be expected to conform to the applicable semiconductor device specification, and to do so under the highest ambient temperature conditions without exceeding the limiting temperatures as specified in table VI. Maximum power dissipation may be estimated from calculations taking into account the mode of operation of the device in the circuit, the worst stress levels applied and the limiting device characteristics as covered by the device specification. Maximum power dissipation may also be estimated by using published data for a particular brand of device, scaling up the losses using a factor calculated from parameter limits applicable to the particular brand of device and specification limits, taking into account the influence of the parameters considered on the losses under the mode of operation in the circuit and allowing for any difference in form factor involved in the way the values are given. Where maximum power dissipation at rated conditions is specified and controlled by the device specification, this value corrected for the stress levels of the application shall be used in the design.

3.5 Environmental requirements.

3.5.1 Shock. The equipment shall withstand the shock test in accordance with MIL-S-901 for grade A, class I, type A, deck-mounted equipment. The equipment shall be hard-mounted, without the use of resilient mounts. The equipment shall operate in parallel with another converter and continually carry load, with the output as specified in MIL-STD-1399, section 300, type III limits during shock test (see 4.7.25).

3.5.2 Vibration. Equipment shall meet the type I vibration requirements as specified in MIL-STD-167-1 up to and including 15 Hz without sway braces. When a higher frequency up to and including 25 Hz is required for a class of ship (see MIL-STD-167-1), the equipment shall meet these requirements with sway braces (see 4.7.26).

3.5.3 Noise. Equipment shall be solidly mounted to the ship structure and shall meet the airborne and structureborne noise requirements as specified in MIL-STD-740-1 and MIL-STD-740-2 (see 4.7.27 and 4.7.28). The limits of airborne noise shall be grade B. The limits of structureborne noise shall be as specified on figure 7.

3.5.4 Electromagnetic interference emission. Equipment shall conform to the electromagnetic interference and susceptibility limits of MIL-STD-461, CE01, CE03, RE02 and RS01 (see 4.7.30).

3.5.5 Ambient temperature. Equipment shall start and operate and conform to all requirements specified herein throughout an ambient temperature range of 0 to 50°C (see 4.7.31).

3.5.6 Cooling.

3.5.6.1 Cooling air. The cooling air for the air cooled units shall be ambient air of a temperature not greater than 50°C. The maximum heat loss to air shall be 22.4 kW at full-rated load of 150 kW (see 4.7.31).

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3.5.6.2 Cooling system failure protection. Protection of the equipment against cooling failure shall be provided by means of automatic de-energizing of the equipment. The input circuit breaker providing power to the equipment shall be shunt tripped in response to overtemperature of equipment components and internal air temperature. Fan failure shall trigger an alarm but shall not trigger shutdown (see 4.7.29).

3.5.7 Humidity conditions. Equipment shall operate with relative humidities ranging up to 95 ± 5 percent by both intermittent and continuous periods, including conditions wherein condensation takes place in and on the equipment cabinets, in the form of both water and frost (see 4.7.34).

3.5.8 Clearance for ventilation and maintenance. Clearance required between the equipment and adjacent equipment or structure shall not be greater than specified in table VII.

TABLE VII. Clearance for ventilation and maintenance.

Bottom	none
Sides or ends	18 inches
Rear	18 inches
Top	12 inches
Front	30 inches

3.5.9 Ship motion. Equipment shall operate when inclined 30 degrees permanently in any direction and when installed on board ships having the following pitch and roll:

	Magnitude	Period
Pitch	10 degrees	5 seconds
Roll	45 degrees	8 seconds

3.5.10 Lifting provision. Each cabinet shall have removable eye-bolts provided on top to permit hoisting and handling. The eye-bolts shall be located to balance the weight of the cabinet as prepared for shipping in the upright position. The lifting attachment shall be of sufficient capacity to carry at least five times the total weight of the cabinet. The cabinet shall remain dripproof when the eye-bolts are removed.

3.6 Reliability. The reliability measure shall be mean-time-between-failures (MTBF) in accordance with MIL-STD-781 and shall be not less than 13,000 hours (see 4.7.38).

3.6.1 Definition of failure. The definition of failure for the purpose of determining the achievement of reliability characteristics shall be:

(a) The input and output performance not in accordance with requirements specified herein.
(b) Malfunction requiring shutdown because of damage or incipient damage.

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3.6.2 Reliability analysis. The reliability analysis shall employ one or more of the prediction methods as specified in MIL-HDBK-217. Failure rates shall be selected as specified in MIL-HDBK-217. When component failure rates are not available as specified in MIL-HDBK-217, failure rates from other sources shall be fully documented and substantiated. For the reliability analysis, the failure rate of each component shall be calculated based on maximum ambient temperature in which the component operates in the equipment. The ambient temperature at each component shall be calculated from the input cooling air temperature which is 50°C. The environmental factor shall be Naval unsheltered as specified in MIL-HDBK-217.

3.6.3 Failure modes and effects analysis (FMEA). The FMEA shall be prepared in accordance with task 101 of MIL-STD-1629.

3.7 Maintainability. The mean time to repair shall be not more than 2 hours. The maximum time to repair shall be not more than 6 hours as calculated in accordance with MIL-STD-470 (see 4.7.37).

3.7.1 Test equipment. The maintenance of the equipment shall require the use of only the following test equipment:

(a) Clamp-on volt ammeter, 0 to 600 amperes and 0 to 600 volts, ac; Nomenclature: CV-633 or AN/USM-33.
(b) Multimeter, 0 to 5000 volts, direct current (dc), 0 to 1000 volts, ac; 0 to 10 amperes, dc to 0 to 30 megohm range; Nomenclature: AN/PSM-4.
(c) Oscilloscope, X-Y, low frequency; Nomenclature: AN/USM-368 (Hewlett Packard Model 130 C or equivalent).
(d) Dc millimeter, 0 to 150 milliampere range; Nomenclature: CV-931-4904003.
(e) Ac voltmeter, 0 to 600 volts range type MR71W600ADVV in accordance with MIL-M-16034 and MIL-M-16034/9.

3.8 Endurance. Equipment shall pass the endurance test specified in 4.7.39.

3.9 Workmanship. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and subassemblies, wiring, welding and brazing, plating, riveting, finishes, machine operations, screw assemblies and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Ambient temperature during inspections. Unless otherwise specified herein, tests shall be conducted under nominal ambient temperature of 25°C. The specific ambient temperature at the time of the test shall be measured and recorded.

4.4 First article inspection. Unless otherwise specified herein, one unit of each power rating shall be subjected to the examination and tests specified in table VIII (see 6.3).

TABLE VIII. First article inspection.

Examination and tests[1/]	Requirement	Inspection
General examination	3.3	4.6, 4.7.1
Insulation resistance	3.3.3.3	4.7.2
Dielectric withstanding voltage	3.3.3.3	4.7.3
Self-protection from input voltage variation and transient	3.2.1.1, 3.2.1.2	4.7.5
Input current harmonics	3.2.1.3	4.7.6
Starting current and starting time	3.2.1.5, 3.2.1.5.1, 3.2.1.6	4.7.7
Voltage and frequency regulation	3.2.2.1, 3.2.2.5	4.7.8
Line drop compensation	3.2.2.2	4.7.9
Voltage adjustment	3.2.2.3	4.7.10
Transient voltage and frequency deviation and recovery	3.2.2.4, 3.2.2.6	4.7.11

See footnote at end of table.

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TABLE VIII. First article inspection - Continued.

Examination and tests	Requirement	Inspection
Amplitude and frequency modulation	3.2.2.7	4.7.12
Output voltage waveform	3.2.2.8	4.7.13
Output phase displacement angle	3.2.2.9	4.7.14
Output voltage balance under balanced load	3.2.2.10	4.7.15
Output voltage balance under unbalanced load	3.2.2.11	4.7.16
Protection for utilization equipment	3.2.7.5	4.7.17
Input voltage change effect	3.2.2.12	4.7.18
Efficiency and no-load input current	3.2.1.7, 3.2.4, 3.2.5	4.7.19
Power rating	3.2.3, 3.2.3.1	4.7.20
Input current balance	3.2.1.4	4.7.21
Overload	3.2.6	4.7.22
Short circuit and fault current	3.2.7.2, 3.2.7.3	4.7.23
Parallel operation	3.2.8, 3.2.8.1, 3.2.8.2, 3.2.8.3	4.7.24
Shock	3.5.1	4.7.25
Vibration	3.5.2	4.7.26
Noise (structureborne)	3.5.3	4.7.27
Noise (airborne)	3.5.3	4.7.28
Cooling system failure protection	3.5.6.2	4.7.29
Electromagnetic interference emission and susceptibility	3.5.4	4.7.30
High temperature operation	3.5.5	4.7.31
Cooling air	3.5.6.1	4.7.31
Inclined operation	3.5.9	4.7.32
Grounding	3.2.1.8	4.7.33
Humidity	3.5.7	4.7.34
Remote start-up and shutdown	3.3.8	4.7.35
Stress level conformation	3.4.5.4	4.7.36
Maintainability demonstration	3.7	4.7.37
Reliability	3.6	4.7.38

1/ These tests may be conducted in any order or concurrently; except that the order of the four last tests shall be: Reliability, dielectric withstanding voltage, stress level conformation, and maintainability.

4.5 Quality conformance inspection. Each unit shall be subjected to the examination and tests specified in table IX (see 6.3).

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TABLE IX. Quality conformance inspection.

Examination and tests 1/	Requirement	Inspection
General examination	3.3	4.6, 4.7.1
Insulation resistance	3.3.3.3	4.7.2
Self-protection from input voltage variation and transient	3.2.1.1	4.7.5
Input current harmonics	3.2.1.3	4.7.6
Voltage and frequency regulation	3.2.2.1, 3.2.2.5	4.7.8
Line drop compensation	3.2.2.2	4.7.9
Voltage adjustment	3.2.2.3	4.7.10
Transient voltage and frequency deviation and recovery	3.2.2.4, 3.2.2.6	4.7.11
Amplitude and frequency modulation	3.2.2.7	4.7.12
Output voltage waveform	3.2.2.8	4.7.13
Output voltage balance under balanced load	3.2.2.10	4.7.15
Output voltage balance under unbalanced load	3.2.2.11	4.7.16
Overload	3.2.6	4.7.22
Short circuit and fault current	3.2.7.2, 3.2.7.3	4.7.23
Parallel operation	3.2.8, 3.2.8.1, 3.2.8.2, 3.2.8.3	4.7.24
Endurance run	3.8	4.7.39

1/ Any of these tests in table IX may be conducted concurrently except that the endurance run shall be conducted after all other tests have been completed, to verify that all requirements have been met.

4.6 General examination. Parts on assemblies and completed units shall be examined to verify that the following requirements have been met:

(a) Sufficient space is provided for these connections and sufficient cable support within the drawers and cabinets, and connections made at the appropriate terminals.
(b) Wherever wires run through holes in metal partitions or chassis, grommets are provided for mechanical protection.
(c) Wires are not bent around sharp corners which may injure the insulation.
(d) Wires are connected by either bolted or soldered connections.
(e) Bolted connections are provided with locking devices.
(f) Both ends of all wires are marked with designations as shown on the accepted drawing.
(g) The equipment pull-out drawers have stops to lock in pull-out position.
(h) Non-identical drawers cannot be interchanged in the equipment cabinet.
(i) Identification and instruction plates are furnished as shown on the accepted drawing.

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4.7 Tests.

4.7.1 Size and weight. Equipment shall be measured and weighed (see 6.3).

4.7.2 Insulation resistance tests. Insulation resistance tests shall be conducted in accordance with method 302 of MIL-STD-202, test condition B.

4.7.3 Dielectric withstanding voltage tests. Dielectric withstanding voltage tests shall be conducted in accordance with method 301 of MIL-STD-202. The rms test voltage shall be twice rated circuit voltage plus 1000 volts. The duration of test shall be 60 seconds. Any evidence of arcing, flashover or punctured insulation shall be interpreted as failure of the test. Corona (visible, audible or odorous) shall not be considered as a failure.

4.7.4 Enclosures. Acceptability of the enclosure shall be determined in accordance with MIL-STD-108 (see 3.3.5).

4.7.5 Self-protection from input voltage variation and transients. The test shall be performed as follows:

(a) Equipment shall be operated at minimum, nominal and maximum input voltages at no-load and rated-load. The input voltage to the equipment shall be raised and lowered beyond the specified operation range for 10 ms. The input voltage, output voltage and output current shall be recorded with an oscillograph.
(b) Equipment shall be operated at minimum, nominal and maximum input voltages to supply rated-load. Interrupt the input power for 30 seconds and then restore to the original condition. The equipment shall be restarted manually. The equipment input voltage, output voltage and output current shall be recorded with an oscillograph.
(c) Equipment shall be operated at nominal input voltage to supply rated-load. Input voltage transients of up to 2500 volts, having a BIL wave shape as shown on figure 1, shall be introduced into the input both line-to-line and line-to-ground. The equipment output shall continue without interruption or malfunction and output waveform shall not deviate from the waveform specified in 3.2.2.8.

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4.7.6 Input current harmonics. The equipment shall be operated at minimum, nominal and maximum input voltage and frequency. The input current harmonics of each input line shall be determined at no-load, 50 percent load and rated-load. Harmonics less than 0.1 percent may be disregarded. The current harmonics shall be measured by means of shunts connected in each input line to the equipment. The fundamental and harmonics of the voltage drops across the shunts shall be determined by a harmonic analyzer.

4.7.7 Starting current and starting time. The minimum, nominal and maximum voltages shall be connected to the equipment input, and the equipment shall be started with no-load at the equipment output. The input starting current and output voltage shall be measured with an oscillograph. The test shall be repeated with full load connected to the equipment. The starting time and starting current shall be determined. An attempt shall be made to energize the equipment when the input voltage phase rotation is in the reverse order (AC-CB-BA). Any damage to the equipment shall be noted.

4.7.8 Voltage and frequency regulation. For first article test, the equipment shall be connected to an input whose voltage and frequency can be varied as specified in table X. After the equipment has been stabilized at load, the output steady state voltage and frequency regulation shall be determined when the test condition (TC) changes as follows:

From	TC2	to	TC36
	TC5	to	TC18
	TC26	to	TC24
	TC33	to	TC34

and when the load changes from 75 percent of rated-balanced-three-phase load to a three-phase unbalanced load, as specified in 3.2.2.11, table III, load condition 1. For quality conformance test, the output steady state voltage and frequency regulation shall be determined when the TC changes from TC2 to TC36 and from TC26 to TC24.

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TABLE X. Conditions of input and output.

Test condition numbers	Input						Output (of rated voltage and frequency)							
	Voltage within the specified range			Frequency within the specified range			Rated load				Power factor			
	Max	Nom	Min	Max	Nom	Min	0	50%	100%	125%	lag. 0.7	lag. 0.8	1.0	lead 0.9
1	X	--	--	--	--	X	X	-	--	--	--	--	-	--
2	--	X	--	--	X	--	X	-	--	--	--	--	-	--
3	--	--	X	X	--	--	X	-	--	--	--	--	-	--
4	X	--	--	--	--	X	-	X	--	--	X	--	-	--
5	--	X	--	--	X	--	-	X	--	--	X	--	-	--
6	--	--	X	X	--	--	-	X	--	--	X	--	-	--
7	X	--	--	--	--	X	-	X	--	--	--	X	-	--
8	X	--	--	--	X	--	-	X	--	--	--	X	-	--
9	--	X	--	--	X	--	-	X	--	--	--	X	-	--
10	--	--	X	--	X	--	-	X	--	--	--	X	-	--
11	--	--	X	X	--	--	-	X	--	--	--	X	-	--
12	X	--	--	--	--	X	-	X	--	--	--	--	X	--
13	X	--	--	--	X	--	-	X	--	--	--	--	X	--
14	--	X	--	--	X	--	-	X	--	--	--	--	X	--
15	--	--	X	X	--	--	-	X	--	--	--	--	X	--
16	--	--	X	--	--	X	-	X	--	--	--	--	X	--
17	X	--	--	--	--	X	-	X	--	--	--	--	-	X
18	--	X	--	--	X	--	-	X	--	--	--	--	-	X
19	--	--	X	X	--	--	-	X	--	--	--	--	-	X
20	X	--	--	--	--	X	-	-	X	--	X	--	-	--
21	--	X	--	--	X	--	-	-	X	--	X	--	-	--
22	--	--	X	X	--	--	-	-	X	--	X	--	-	--
23	X	--	--	--	--	X	-	-	X	--	--	X	-	--
24	X	--	--	--	X	--	-	-	X	--	--	X	-	--
25	--	X	--	--	X	--	-	-	X	--	--	X	-	--
26	--	--	X	--	X	--	-	-	X	--	--	X	-	--
27	--	--	X	X	--	--	-	-	X	--	--	X	-	--
28	X	--	--	--	--	X	-	-	X	--	--	--	X	--
29	X	--	--	--	X	--	-	-	X	--	--	--	X	--
30	--	X	--	--	X	--	-	-	X	--	--	--	X	--
31	--	--	X	X	--	--	-	-	X	--	--	--	X	--
32	--	--	X	--	X	--	-	-	X	--	--	--	X	--
33	--	X	--	X	--	X	-	-	X	--	--	--	X	--
34	--	X	--	X	--	--	-	-	X	--	--	--	X	--
35	--	X	--	--	X	--	-	-	--	X	X	--	-	--
36	--	X	--	--	X	--	-	-	--	X	--	X	-	--
37	--	X	--	--	X	--	-	-	--	X	--	--	X	--

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4.7.9 Line drop compensation. The unit shall be connected to rated-load and the load compensation shall be adjusted to obtain a 7 percent boost (31.5 volts) in output voltage. The load shall be reduced to half-load and to no-load. The average of the three output phase voltages shall vary linearly from full load to no-load within plus or minus 0.5 percent (plus or minus 2.25 volts). This test shall be performed at minimum, nominal and maximum input voltage conditions as specified in 3.2.1.

4.7.10 Voltage adjustment. After the equipment has been stabilized at rated-load, the output voltages at the equipment output terminals shall be adjusted over their full ranges, using the voltage adjustment devices. For the first article test, the test shall be conducted for test conditions 2, 23, 24, 25, and 26 as specified in table X. For quality conformance test, the test shall be conducted for test conditions 2 and 25 as specified in table X.

4.7.11 Transient voltage and frequency deviation and recovery. With the equipment operating with nominal input, a balanced 3-phase load of 0.8 power factor shall be changed in one step for the following conditions:

(a) From no-load to half-load
(b) From half-load to rated-load
(c) From rated-load to half-load
(d) From three-fourths load to one-fourth load

Oscilloscope pictures shall be taken of the load application and removal to determine the voltage transient and recovery time. This test shall be conducted at minimum, nominal and maximum input conditions as specified in 3.2.1. The frequency transient and recovery test shall be performed the same way except that load application and removal shall be from no-load to rated-load and from rated-load to no-load in one step, respectively.

4.7.11.1 Steady-state nonlinear load. The equipment shall be operated at nominal input voltage and a balanced nonlinear load consisting of a multiphase rectifier/filter with a dc load. The test shall be run with dc loads equal to 25, 50 and 70 percent of the rated equipment kW load applied to the rectifier/filter. A second series of tests shall be run with loads consisting of equal 0.8 (lagging) PF and nonlinear loads (as described herein) at 25, 50 and 70 percent of rated equipment kW load. The output voltage waveform shall remain within the limits as specified in 3.2.2.8; output voltage regulation shall be in accordance with 3.2.2.1 and the voltage amplitude modulation shall be in accordance with 3.2.2.7.

4.7.12 Amplitude and frequency modulation. For first article and quality conformance tests, the amplitude and frequency modulation of the equipment shall be determined at the equipment output terminals for the conditions specified below:

(a) For the changes specified in 4.7.8.
(b) When the output voltage is adjusted from 427.5 to 472.5 volts and from 472.5 to 427.5 volts.

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4.7.13 Output voltage waveform. The harmonic content and output voltage spikes shall be determined. For first article, this test shall be performed with the equipment operating under the test conditions 2, 25, and 30 as specified in table X. For quality conformance test, this test shall be performed under the test conditions 2 and 25 as specified in table X.

4.7.14 Output phase displacement angle. The equipment output displacement angle shall be measured at the output terminals. For first article and quality conformance tests, this test shall be performed for test conditions 2 and 25, as specified in table X.

4.7.15 Output voltage balance under balanced load. After the equipment has been stabilized at rated-load, the output voltage of each phase at the equipment output terminal shall be recorded. For first article test, this test shall be performed for test conditions 1, 2, 3, 7, 9, 11, 23, 25, and 27 as specified in table X. For quality conformance test, this test shall be performed for test conditions 2, 14 and 25 as specified in table X.

4.7.16 Output voltage balance under unbalanced load. Equipment shall be operated with nominal input, and a 3-phase, 0.7 lagging power factor load shall be applied across the equipment output phases to have the lines carry current of 75 percent of rated current. A single phase 0.7 lagging power factor load shall be applied across the equipment output phase A-B to have line A carry an additional current of 15 percent of rated current. The output voltage of all phases shall be recorded at the equipment output terminals. The test shall also be conducted using a single phase 0.7 lagging power factor load across equipment output phase B-C and then C-A. The equipment shall be operated at nominal input. A single phase, 0.7 lagging power factor load shall be applied across equipment output phases A-B to have line A carry a current of 15 percent of the rated current. The output voltages of all phases shall be recorded. The test shall also be conducted using a single phase 0.7 lagging power factor load across equipment output phases B-C and then C-A. The above tests shall be repeated for 0.8 lagging power factor load.

4.7.17 Protection for utilization equipment. Equipment shall be operated at nominal input at no-load. Voltage signals shall be injected into equipment sensing circuits to simulate the condition of equipment output overvoltage and undervoltage, as specified in table IV. The output voltage and the time required to shut down the equipment in the case of overvoltage shall be recorded by oscilloscope or oscillograph.

4.7.18 Input voltage change effect. Equipment shall be operated within the input specified voltage and frequency ranges, and the input voltage shall be increased or decreased to the specified amount (see 3.2.2.12) instantaneously. The input and output voltages shall be recorded by oscillograph to determine the voltage transient and recovery time. The test shall be performed at no-load and rated-load, 0.8 power factor.

4.7.19 Efficiency and no-load input current. The equipment shall be loaded with 0.8 power factor loads for this test. Meters shall be inserted in the input and output for reading voltage, current and wattage. For first article test,

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readings shall be made for test conditions 2, 25, 30 and 32 as specified in table X. For quality conformance test, readings shall be made for test conditions 2 and 25 as specified in table X. Efficiency shall be calculated by means of the following formula:

$$\text{Overall percent efficiency} = \frac{\text{Output power}}{\text{Input power}} \times 100$$

4.7.20 Rating. The equipment shall be operated at nominal input for 2 hours each at the following loads:

(a) Rated-load, 0.7 power factor lagging (105 kW)
(b) Rated-load, 0.8 power factor lagging (120 kW)
(c) Rated-load, unity power factor (150 kW)
(d) Rated-load, 0.9 power factor leading (135 kW)

Input power, output power, output current, output frequency and output voltage shall be recorded at the beginning and at the end of each 2-hour load run.

4.7.21 Input current balance. The input voltage to the equipment shall be nominal with 3 percent unbalance among the three phases. The equipment shall be operated at 15 percent load, 50 percent load, and rated-load at rated output voltage and frequency. The input current of each line to the equipment shall be measured with indicating meter having initial accuracy of 1 percent or better. This test shall be repeated with the 3-phase voltages set at the nominal value and balanced within 4 volts.

4.7.22 Overload. The equipment shall be operated at minimum input voltage and rated output and then overloaded to 125 percent of rated-load at rated output voltage, 0.8 power factor for more than 30 minutes. The output voltages, output currents and output amplitude modulation shall be determined by an oscillograph.

4.7.23 Short circuit and fault current. A 3-phase and single phase short circuit fault shall be applied to the equipment output terminals and at the end of 10, 75 and 150 foot cables. These tests shall be conducted with nominal input voltage. The fault current and the time required to shut down the equipment shall be recorded with an oscillograph and also with an oscilloscope. Output fault current simulating 210 and 300 percent overloads as specified in 3.2.7.2 shall be simulated at nominal input voltage. The time the unit provides this fault current overload shall be recorded by an oscilloscope, oscillograph or timer.

4.7.24 Parallel operation. For first article test, not less than two units of the same kW rating shall be operated in parallel. For quality conformance test, two like units of the same kW rating shall be operated in parallel. When units are operated in parallel, the following tests shall be performed: One of the units shall be supplied from one input power source and the other from a different input power source, the sources shall be at different voltages but not exceeding 2 percent. After the units have been automatically synchronized to operate in parallel, no further adjustment shall be made for the remainder of

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this test. The total load of the paralleled units shall be varied from no-load to 90 percent of the combined kW and kVA rating of the units at 0.8 power factor in 25 percent load steps and back to no-load in the same manner. The units shall operate for not less than 1 hour at each load step. The difference in kW and kVA supplied by each unit to the load shall be recorded and computed. For each load step the parallel test shall be conducted at the maximum, nominal and minimum input voltage for 20 minutes each in the specified range. Simulated internal fault such as an oscillator or inverter module loss or door interlock opening shall be applied to one of the paralleled units to demonstrate selective tripping. Selective tripping shall also be simulated by removal of input power to one of the units; the equipment output voltage shall be recorded by an oscillograph or oscilloscope photo. After completion of the selective tripping test, the units shall operate in parallel at 20 percent load for 16 hours continuously.

4.7.25 Shock. Two like frequency changers shall be connected in parallel output and, while both are operating, one unit shall be subjected to high-impact medium weight shock tests in accordance with MIL-S-901 and as specified in 4.7.25.1.

4.7.25.1 Test features. Test features shall be as follows:

(a) Type A (for principal units), deck-mounted equipment.
(b) Grade: A.
(c) Equipment class: I.
(d) Load applied: 20 percent of rated-load, both units.
(e) Mounting: hard-mounting.
(f) Prior to and following the shock test, the units shall be tested as specified in MIL-STD-1399, section 300, type III limits.
(g) Definition of failure to perform principal functions shall be as follows:
    (1) Breaking of any parts, including mounting bolts. Minor chipping of parts such as plastic knobs and cases and minor distortion of parts will be permitted where such chipping or distortion cannot in any manner impair operation of the equipment as specified.
    (2) Appreciable distortion of any parts, including enclosure and framework.
    (3) A value of insulation resistance lower than that permitted by this specification (see 3.3.3.3).
    (4) Failure to pass visual examination. The equipment shall be carefully examined after removing all removable panels, doors and drawers to ascertain any mechanical damage. If necessary, partial disassembly shall be performed to aid in determining any possible damage.
    (5) Failure to continually carry rated-load during shock test and to perform electrical test following shock. No adjustment or replacement of damaged parts will be permitted after the shock test and prior to the tests specified in (h).
(h) On completion of the shock tests, the units shall be tested for insulation resistance and output performance as specified in MIL-STD-1399, section 300, type III limits.

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4.7.26 Vibration. Vibration tests shall be conducted in accordance with the type I vibration requirements of MIL-STD-167-1 up to and including 15 Hz. The equipment shall be tested with 20 percent or more rated-load. Sway braces shall not be used unless the required frequency range is up to and including 25 Hz.

4.7.26.1 Failure. The definition of failure as specified in 4.7.25.1(g) shall also apply to vibration tests.

4.7.27 Structureborne noise. The structureborne noise shall be determined in accordance with MIL-STD-740-2. For the noise tests, the equipment shall be operated at full load 0-8 PF, and maximum input voltage. Measurements shall be made in three orthogonal directions above every isolation mount. The power sum is defined as:

$$(\text{Power sum adB})_{\kappa} = \left[\left(10 \log_{10} \frac{1}{n} \sum_{i=1}^{n} 10^{(a_i/10)}\right) \right]_{\kappa}$$

Where:

a_i - acceleration level in adB (referred to 10^{-3}cm/sec^2 above the ith isolation mount).

n - the number of measurement points in a given direction (number of isolation mounts).

κ - direction - vertical, longitudinal or transverse.

The power sum as measured for any of the three directions shall not exceed the limits of figure 7 (see 6.3).

4.7.28 Airborne noise. The airborne noise shall be determined in accordance with MIL-STD-740-1. Load conditions during tests shall be as specified for structureborne noise tests (see 4.7.27 and 6.3).

4.7.29 Cooling system failure protection test. The equipment shall be operated at nominal voltage input and at rated-load. The cooling fan circuits shall be interrupted to simulate fan failures. The equipment input voltage and output voltage shall be observed with an oscilloscope.

4.7.30 Electromagnetic interference emission and susceptibility. An electromagnetic interference emission and susceptibility test shall be conducted in accordance with MIL-STD-462. Tests shall be conducted on each input and output conductor of the equipment. Data for both the broad band and narrow band of the conducted emission levels shall be taken (see 6.3).

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4.7.31 High temperature operation. High temperature operation tests shall be conducted at an ambient temperature of 50°C. The test methods to be employed and the precautions to be observed shall be as specified in 4.7.31.1 through 4.7.31.5.

4.7.31.1 Assembly and mounting of equipment. High temperature tests shall be made only with the equipment completely assembled and mounted in the manner for which it is designed. Barriers shall be placed adjacent to the enclosure to simulate shipboard space restrictions. The minimum temperature measurement points shall be as follows:

(a) Power semiconductor
(b) Capacitors
(c) Power transformer and reactor coil windings
(d) Ambient temperature
(e) Cooling air inlet and outlet

4.7.31.2 Method of temperature measurements. The methods of temperature measurement of ambient and parts or subassemblies shall be in accordance with MIL-E-917 except that a thermometer shall not be used.

4.7.31.3 Method of loading. The equipment shall be operated at test conditions 23, 25 and 27 as specified in table X. The temperatures shall reach the steady final value as specified in 4.7.31.4 for each input voltage and for each load condition. Temperature rises shall be in accordance with MIL-E-917.

4.7.31.4 Duration of test. The consecutive temperature measurement of each part or module shall be recorded not more than 3 minutes apart. The temperature rise test shall be continued until the temperature rise of parts or modules that can be observed or recorded during test have attained a steady final value. Steady final value is that value of temperature which will remain at a 2°C band for 2 hours, while the ambient temperature is maintained at 50°C.

4.7.31.5 High temperature tests. After the equipment has attained a steady final value as specified in 4.7.31.4, the following tests shall be conducted at the ambient temperature of 50°C:

(a) Voltage and frequency regulations tests as specified in 4.7.8.
(b) Transient voltage and frequency deviation and recovery test as specified in 4.7.11.
(c) Amplitude and frequency modulation tests as specified in 4.7.12.
(d) Output voltage waveform test as specified in 4.7.13.
(e) Short circuit protection test as specified in 4.7.23.
(f) Inclination test as specified in 4.7.32.

4.7.32 Inclined operation. The equipment shall be tested for inclined operation at 30 degrees from horizontal. The equipment shall be run at rated-load for not less than 1 hour at each of the following positions at an ambient temperature of 50°C.

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(a) 30 degrees forward.
(b) 30 degrees backward.
(c) 30 degrees to left side.
(d) 30 degrees to the right side.

During the progress of this test, it shall be ascertained that there is no apparent overheating of any drawers, modules or parts; dropping out of circuit breakers; or malfunction of operation of the equipment.

4.7.33 Grounding. The test shall be performed as follows: The equipment shall operate at minimum, nominal and maximum input voltage and at no-load and rated output load. Each input and output terminal shall be grounded in turn to ground as follows:

(a) Connect each terminal to ground through a 100,000 ohm resistance.
(b) Momentarily ground each terminal in turn five times by shorting one of the 100,000-ohm resistors in (a). Remove that short and then short the other 100,000-ohm resistor. Also ground each terminal for 10 seconds and then remove the short.

4.7.34 Humidity. The equipment shall be tested in accordance with MIL-STD-810, modified so that the maximum chamber temperature is 50°C.

4.7.35 Remote start-up and shutdown. The equipment shall be operated at nominal input voltage. The control of the equipment shall be from a remote control panel equivalent which is connected to the equipment with a cable having an equivalent resistance of a 500-foot type LSMSCU cable in accordance with MIL-C-24643 and MIL-C-24643/18. Equipment operation from the remote control panel shall be demonstrated.

4.7.36 Stress level conformation. The electrical stresses of the parts in the equipment shall be tested to conform to the derated stress limits of 3.4.5.4 and NAVSEA 0967-LP-597-1011. Stress level shall be tested on representative parts as approved by NAVSEA.

4.7.37 Maintainability demonstration. The equipment shall be subjected to maintainability demonstration tests in accordance with MIL-STD-471, method I, test B for mean corrective maintenance time (Mct) (15 samples). Maintenance tasks of the maintainability demonstration tests shall be performed by military or civil service personnel of the level and training available during operational use. After completion of each maintenance action, the equipment shall be operated for 15 minutes at rated-load. The time measured for each corrective maintenance task shall consist of the sum of the following steps: Fault location, fault correction, adjustment, if any, and check-out, including the consulting of technical manuals and other allowed aids during this time.

4.7.38 Reliability. After completion of all other tests, a 2000-hour reliability test shall be accomplished in accordance with MIL-STD-781 and MIL-HDBK-781. The test shall be performed with the unit at 50°C only.

4.7.39 Endurance test. A continuous 200-hour endurance run shall be conducted on the equipment. The test shall be as follows:

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(a) Ambient temperature cycling. Operate equipment at 50 ± 5°C for 15 hours. Then reduce temperature to ambient of 25 ± 5°C for 5 hours. Repeat the cycle 10 times continuously.

(b) Output load cycling. When the equipment is operating at 50 ± 5°C, the rated-load shall be applied continuously for 13 hours followed by 2 hours at no-load. When the equipment is operating at 25 ± 5°C, the rated-load shall be applied continuously for 3 hours followed by 2 hours at no-load. The contractor may, for the purpose of energy management, schedule the rated-load operating hours and power down so as to avoid the normal peak utility load operating times.

(c) Output voltage and frequency. 450 volts ac nominal, 400 Hz.

(d) Input voltage and frequency. 440 volts ac nominal, 60 Hz.

(e) Data. The input current, input voltages, input frequency, output current, output voltage, output frequency and output power, test chamber ambient temperature, outside chamber ambient temperature and operating time, shall be taken automatically every 15 minutes. Results shall be verified every 50 hours by test personnel.

(f) If a failure should occur during the first 130 hours of operation, the cause of failure shall be recorded, corrected and the test continued. The equipment shall operate the final 70 hours without failure. If any failure occurs, the final 70 hours shall be repeated without failure.

4.8 Inspection of packaging. Sample packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.8.)

5.1 General.

5.1.1 Navy shipboard stowage fire-retardant requirements.

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	-	Type II - weather resistant. Category 1 - general use.
Level C	-	Type I - non-weather resistant. Category 1 - general use.

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(b) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation, packing and marking. The equipment shall be preserved level A, C, or commercial, packed level A, B, or C or commercial as specified (see 6.2) and marked in accordance with MIL-E-17555.

5.3 Special requirements for critical close tolerance equipment. Special preservation, packing, and marking in addition to any preservation and packing requirements specified by the product specification and packing requirements code, equipment which are identified to MIL-STD-740-1 and MIL-STD-740-2 shall be protected against damage resulting from environmental conditions, multiple handling and the hazards of transportation (rough handling, shock, vibration, and so forth). Shipping containers or method of packing utilizing shock or vibration mitigation systems shall only use mounts which have captive features. Unit packs and shipping shall be marked as specified (see 6.2).

5.3.1 Special marking unit packs and shipping containers and unpacked shipments shall be marked with the following:

"CRITICAL, CLOSE TOLERANCE OPERATING EQUIPMENT
HANDLE WITH CARE
DO NOT DROP OR SUBJECT TO SHOCK OR JARS"

Markings shall be stencilled, red color, and applied on two sides and both ends of the container for shipment. Letters shall be a minimum 1-1/2 inch high, except for small containers with insufficient space, in which case letters shall be of such size as to be legible. In addition, arrows and the word, "UP", center of balance, sling or lifting point markings as indicated in MIL-E-17555 shall apply.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Solid state frequency changers covered by this specification are intended to convert Navy shipboard 60 Hz electrical power into 400 Hz electrical power for surface ships.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Whether first article inspection is required (see 3.1).
(d) When semiconductor devices are other than silicon (see 3.4.5).
(e) When fire-retardant treated materials are not required (see 5.1.1(a)).

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(f) Levels of preservation, packing, and marking required (see 5.2).
(g) Special marking (see 5.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
3.3.1	DI-E-7031	Drawings, engineering and associated lists	Level 3
3.6.3	DI-R-7085	Failure mode, effects, and criticality analysis report	----
4.1.1	DI-R-7064	Calibration system description	----
4.4	DI-T-4901	First article inspection procedure	----
4.4	DI-T-4902	First article inspection report	----
4.5, 4.7.1	DI-T-5329	Inspection and test reports	----
4.7.27	DI-HFAC-80274	Equipment structureborne vibration acceleration measurements test report	----
4.7.28	DI-HFAC-80272	Equipment airborne sound measurements test reports	----
4.7.30	DI-EMCS-80200	Electromagnetic interference test report	----

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

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6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first _____ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.5. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Definitions.

6.6.1 Equipment. Equipment is defined as a solid state frequency changer of the power rating defined by purchase specification.

6.6.2 Unit. Unit is defined as a single, individual solid state frequency changer housed in one enclosure.

6.6.3 Cabinet. Cabinet is defined as a number of drawers physically joined together to perform a specific function.

6.6.4 Drawer. Drawer is defined as two or more parts or removable modules in one drawer. A drawer is a portion of a cabinet.

6.6.5 Module. Module is defined as two or more parts to form an entity as part of a drawer. Module can be replaceable as a whole, but having a part or parts which are individually replaceable.

6.6.6 Part. Part is defined as one piece, or two or more pieces, joined together which are not normally subjected to disassembly, without destruction of intended use. Parts are the lowest level of breakdown.

6.7 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.7.1 When ordering spare parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

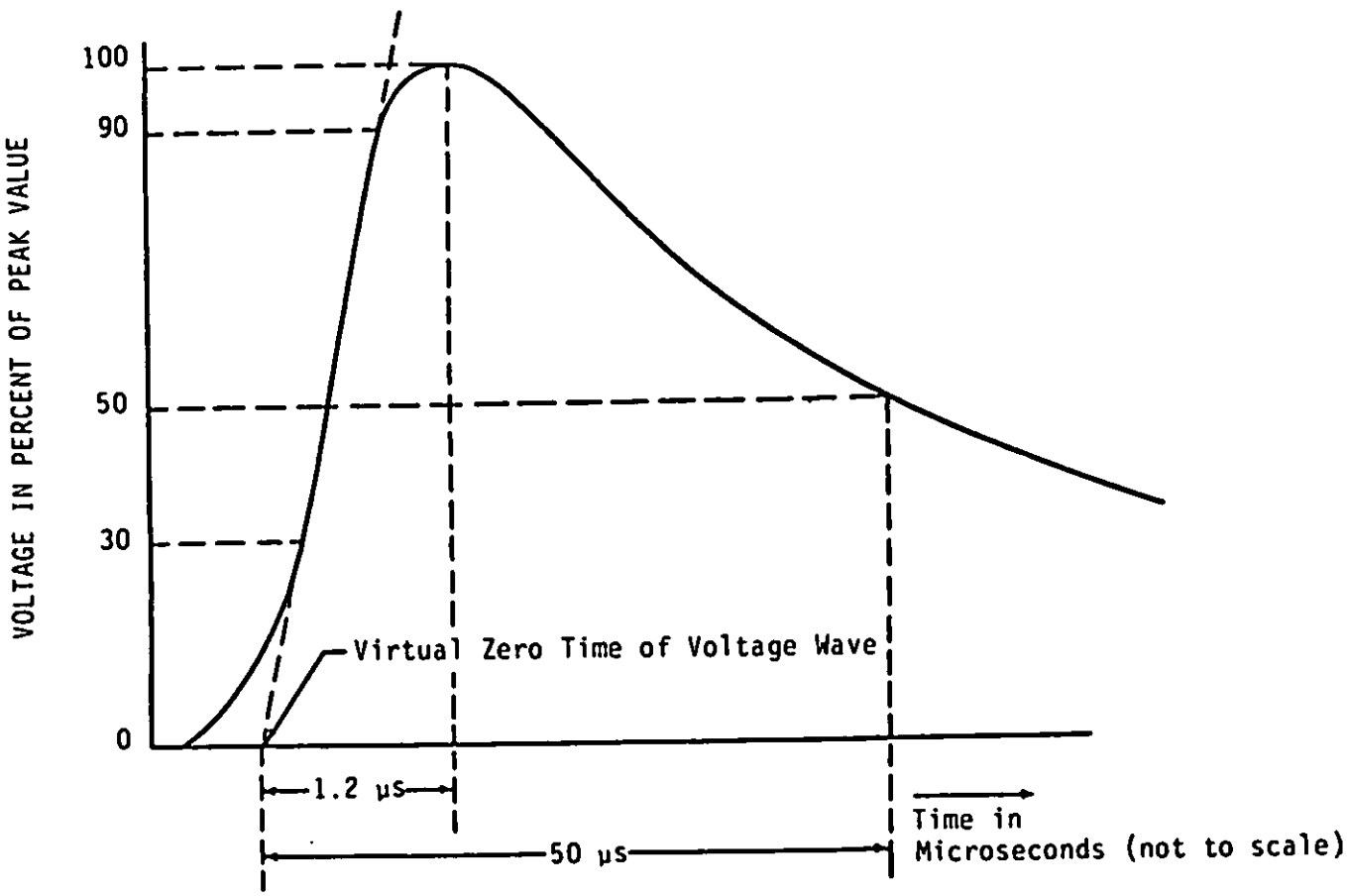
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6.9 Subject term (key word) listing.

Harmonics
Heat dissipaters
Input voltage
Line drop compensation
Output voltage
Semiconductor device

Preparing activity:
Navy - SH
(Project 6130-N295)

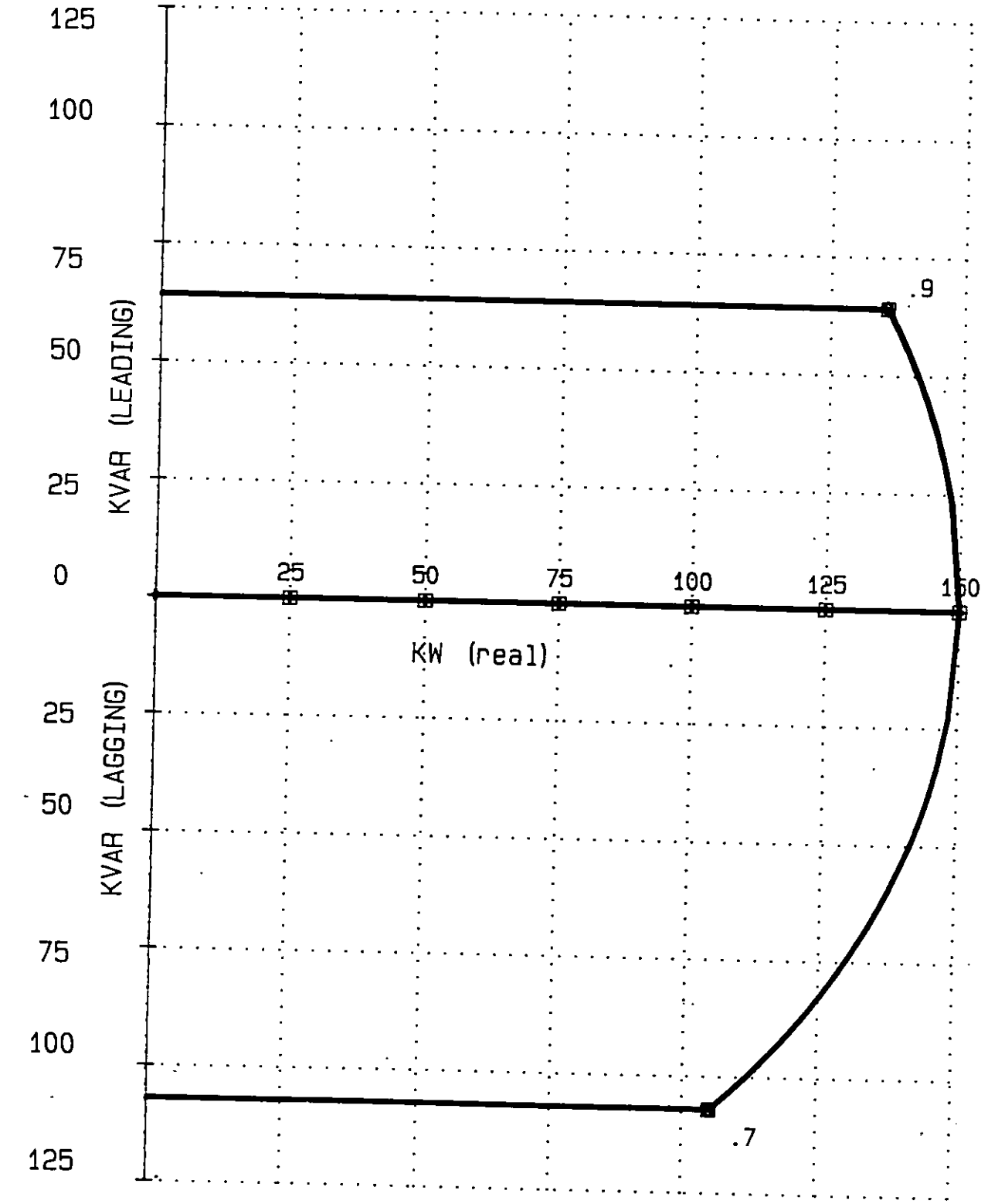
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SH 13231997

FIGURE 1. Spike voltage (short time transient) wave shape.

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SH 13231998

FIGURE 2. Continuous power rating.

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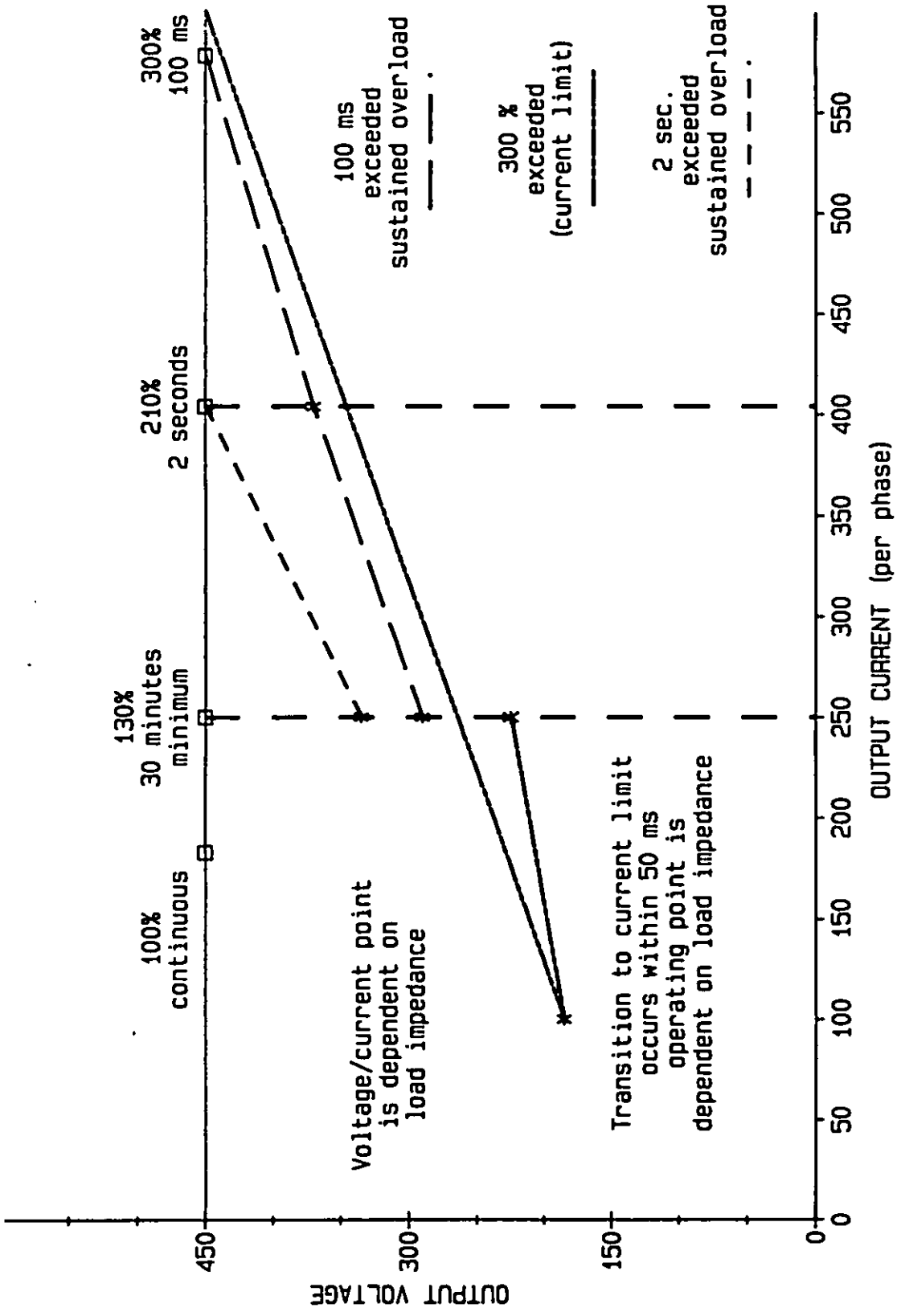


FIGURE 3. Fault current performance, current voltage domain.

SH 13231999

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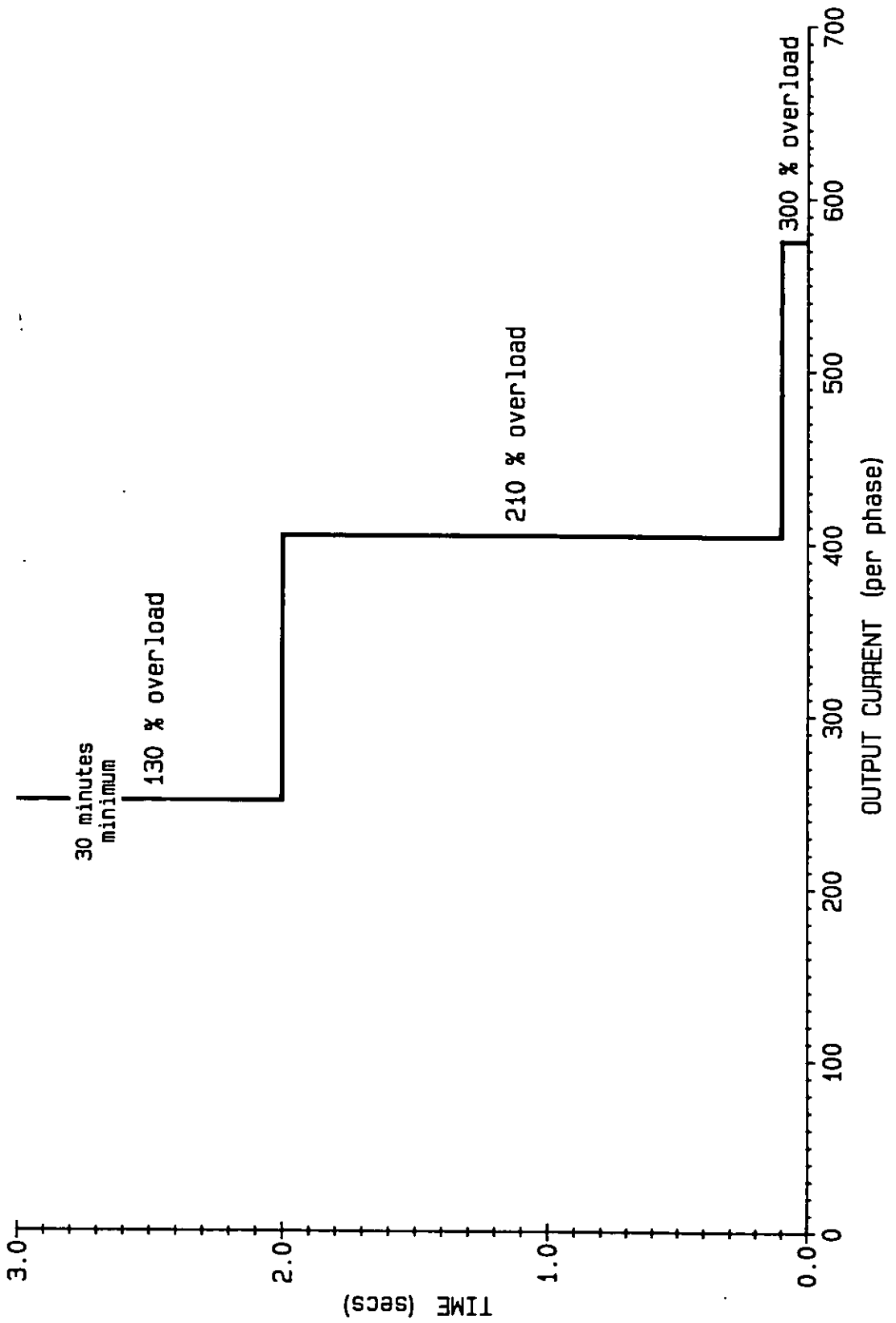
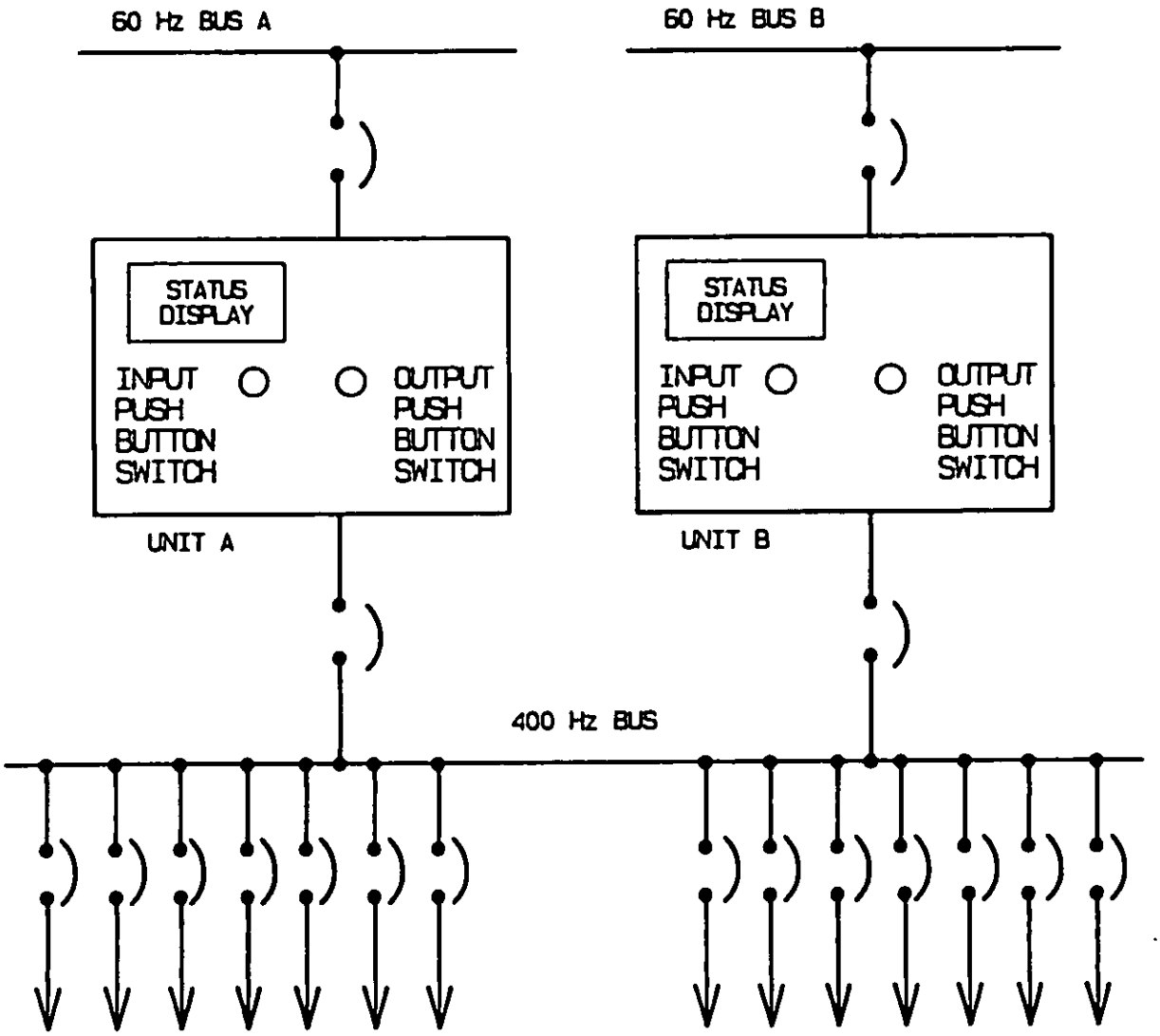


FIGURE 4. Fault current performance, current-time domain.

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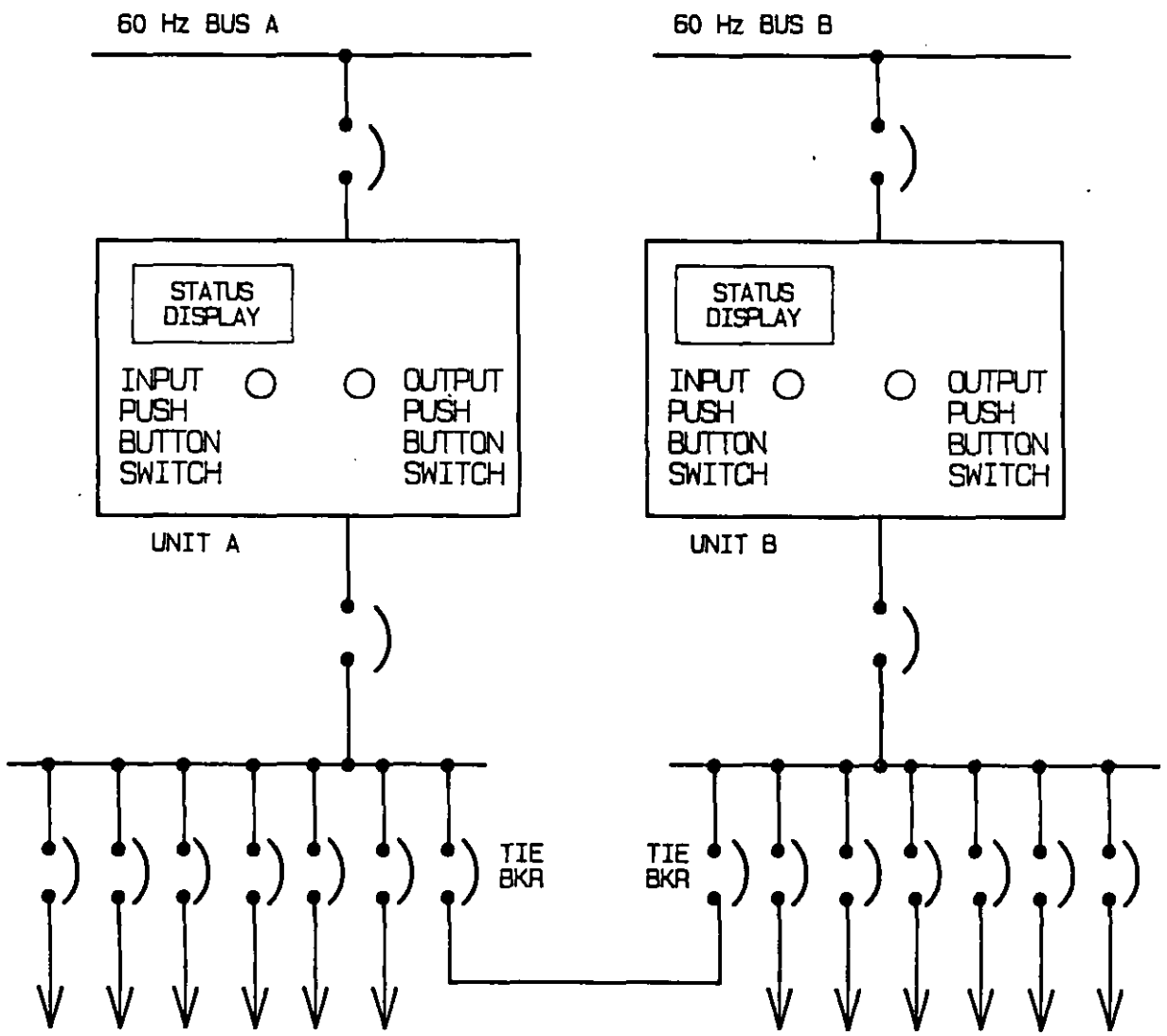


NOTE: Shown for two units only, but same arrangement applicable for up to six units.

SH 13232001

FIGURE 5. Parallel operation - common bus.

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NOTE: Shown for two units only, but same arrangement applicable for up to six units.

SH 13232002

FIGURE 6. Parallel operation - split bus.

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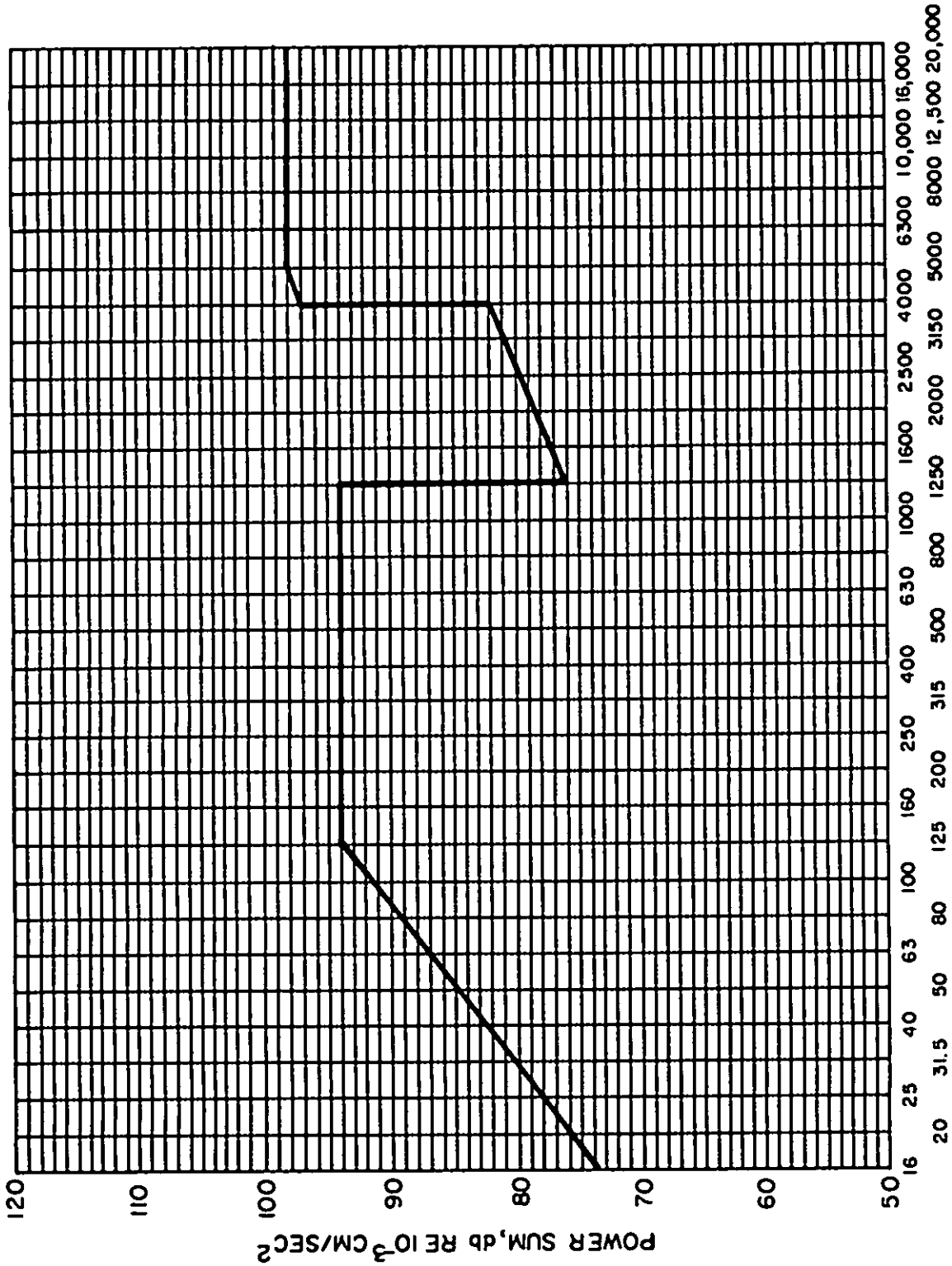


FIGURE 7. Structureborne noise plot.

SH 13232003

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-24713(SH)	FREQUENCY CHANGER, STATIC, AIR COOLED (NAVAL SHIPBOARD)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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